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Bioblitzes provides valuable biodiversity data

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Bioblitzes provides valuable biodiversity data

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Abstract

Bioblitzes are increasingly popular participatory science events aimed at documenting as many species as possible over a pre-defined period. They enhance community engagement and increase scientific literacy, while also contributing immense amounts of biodiversity data in a relatively short time span. Yet, the value of the biodiversity data generated is rarely quantified. In this study, we used the DeLuca Bioblitz as a case study to develop a biodiversity utility framework and a list of biodiversity metrics to evaluate the biodiversity data contributions from bioblitzes. We calculated summary statistics, including the total number of observations, number of observers, and number of species from the DeLuca Bioblitz. We compared these summary statistics to random polygons of the same area across Florida, to evaluate the impacts of bioblitz efforts compared to participatory science generally. Bioblitzes tend to generate more biodiversity data when compared to other random polygons of the same size throughout the state of Florida. Importantly, Bioblitz data was collected over a 24-hour period for 4 years, yet it outpaced biodiversity contributions in random polygons with data collected over multiple years on iNaturalist. The bioblitz in our case study also generated key records of new invasive species, new continental records, and records of imperiled species in Florida. Together, these data can help land managers and conservation professionals collect robust data, over a short duration, to create baseline species inventories and document species of concern for both management and conservation.

Keywords: community science; citizen science; adaptive management; bioblitz; iNaturalist

Introduction

Site-level species inventories, such as a list of possible species in a certain property or park, often serve as baseline data for natural resource managers (Maes et al. 2015; Katzer et al. 2025). This information is essential for natural resource managers to prepare management plans and monitor ecological changes over time, particularly as species shift their ranges (Weiskopf et al. 2020) and local extinction and colonization events become increasingly common (Cowie et al. 2022; Storch et al. 2022). However, gathering such data can be time-consuming and costly; therefore, inventories can be difficult to maintain (Katzer et al. 2025). Advances in technology have given rise to many long-term monitoring platforms, including participatory citizen science, sometimes referred to as community science or citizen science. Participatory citizen science is increasingly contributing to biodiversity research (Fauzia et al. 2023; Mason et al. 2025a), conservation (Hsu et al. 2025), and natural resource management (McKinley et al. 2017; Mason et al. 2025b), for example by quantifying species distributions (Zulian et al. 2021) or detecting biological invasions (Larson et al. 2020). The bulk of all biodiversity data in the world now comes from platforms such as eBird and iNaturalist—upwards of 80% of all global biodiversity data since 2010 (Callaghan et al. 2020).

Bioblitzes are one mechanism in which participants are encouraged to contribute data through participatory science. A bioblitz has many definitions (Meeus et al. 2023), but in general, it involves a set time period where the goal is to document as many species as possible within a pre-defined geographic region (Meeus et al. 2023). This can range from a single park to large geographic regions. One particularly well-known example is the City Nature Challenge — a four-day bioblitz organized annually by the California Academy of Sciences — where cities

across the world participate in a friendly competition to get the most observations, species, and people using iNaturalist (Palma et al. 2024; Kishimoto & Kobori 2021; Sakurai et al. 2022). Bioblitzes are useful for engaging the public in conservation and creating a sense of personal involvement in environmental research (Roger & Klistorner 2016). They are also useful for generating large amounts of data in a given location (Meeus et al. 2021, 2023). The City Nature Challenge alone provides the most data of any week to the iNaturalist platform each year (Meeus et al. 2023; Palma et al. 2024). Despite their global popularity, most bioblitzes occur at localized scales, and consequently their scientific value has been less frequently quantified.

Bioblitzes can have a variety of outcomes, including building community, increasing engagement in science, incentivizing continued participation on platforms like iNaturalist, and generating biodiversity data (Meeus et al. 2021, 2023; Pollock et al. 2015; Postles and Bartlett 2015). They also contribute to species inventories, but less is known about other types of biodiversity data they help generate. Bioblitzes may be beneficial for biodiversity monitoring by detecting rare and cryptic species, new invasive species to a region, range shifts in imperiled species, or help filling gaps in a species' geographic distribution. Therefore, it is critical to uncover the varied biodiversity utility that bioblitzes offer beyond simple species inventories for both land managers and conservation practitioners alike.

Here, we evaluate the value of a bioblitz to generate biodiversity data, using the DeLuca Preserve, FL, USA, as a case study. First, we provide a replicable framework that can be expanded outside of our study site, with the aim of helping natural resource managers leverage bioblitzes to enhance land stewardship goals and advance biodiversity data collection. We then

use this framework to illustrate the scientific value of a bioblitz as a mechanism for generating biodiversity data, including baseline species inventories, novel species records, and indicators of regional biodiversity utility (Figure 1). Our work highlights how concentrated, short-term efforts can uncover locally novel records and contribute to site-level species inventories in ways that complement broader-scale biodiversity monitoring.

Bioblitz biodiversity utility framework

We present a conceptual framework for evaluating the biodiversity value of bioblitzes by organizing metrics along two dimensions: their biodiversity utility for management and monitoring, and the analytical complexity required to calculate them (see Figure 1). The framework distinguishes site-level metrics that characterize biodiversity patterns within the site defined for the bioblitz (this includes, total observations, number of species, number of observers, and new species) from regional-level metrics that contextualize bioblitz observations relative to broader regional biodiversity data (this includes, relative frequency, relative rarity, and relative diversity of species, sampling intensity, and nearest external observation). Together, these metrics capture complementary dimensions of biodiversity value, ranging from baseline inventory completeness to the contribution of short-term, concentrated effort to regional biodiversity knowledge. Because we separate site-level and regional-level metrics, this framework provides a flexible tool that natural resource managers can adapt to evaluate the biodiversity outcomes of bioblitzes across different regions, taxa, and management objectives.

Methods

Study area

The DeLuca Preserve (27.67, -80.95) is situated in Osceola County in central Florida. The DeLuca Preserve is located near Yeehaw Junction, FL, a census-designated place in southern Osceola County about three hours south of the main University of Florida campus in Gainesville, FL, and about three hours north of the Miami metropolitan area in South Florida. The preserve was donated to the University of Florida by Elisabeth DeLuca in 2020 to place the land in permanent conservation (in collaboration with Ducks Unlimited) and to provide high-quality habitat for student research. DeLuca Preserve is ~11,000 hectares and consists of pine flatwoods, Florida scrub, hardwood hammocks, forested and emergent wetlands, imperiled dry prairie, former citrus groves, and working cattle ranchland. The preserve hosts several state and federally listed species, such as the Florida grasshopper sparrow (*Ammodramus savannarum floridanus*), red-cockaded woodpecker (*Leuconotopicus borealis*), gopher tortoise (*Gopherus polyphemus*), and Florida panther (*Puma concolor couguar*). The area is now managed by the University of Florida Institute for Food and Agricultural Sciences (UF/IFAS) with varied management activities including active cattle ranching, hunting, prescribed burning in the pine flatwoods and pastures.

The DeLuca Bioblitz

The DeLuca Bioblitz was initiated by the University of Florida Institute of Food and Agricultural Sciences (UF/IFAS) to promote public engagement with the newly established preserve while showcasing a conservation focal area of the Everglades Headwaters National Wildlife Refuge and Conservation Area. This annual event is designed as a free-ranging bioblitz, meaning participants are free to sample taxa and habitats of interest without any restrictions. However, we

note that in the early years, there were some concerted efforts to obtain baseline biodiversity metrics with limited resources and people, which meant targeting specific portions or habitats of the property.

To date, the DeLuca Bioblitz has been running for four years and is expected to continue in future years. The dates of the previous bioblitzes were as follows: 26 February 2022, 24 February 2023, 1 March 2024, and 7 March 2025. Over the course of these four years, participants have generated 13,161 total observations by 203 unique observers. With annual increases in total observations and unique observers, the DeLuca Bioblitz is positioned to play a long-term role in documenting local biodiversity trends and fostering awareness of the ecological significance of the DeLuca Preserve.

iNaturalist

iNaturalist is a citizen science platform where users can post and identify photographic or audio observations of any organism. It is accessible via the iNaturalist website or through their mobile application. As of October of 2025, iNaturalist has over 3 million observers and more than 200 million observations. After an observation has been uploaded with a species identification on iNaturalist, other users can agree or disagree with the identification. Once an observation reaches a two-thirds community consensus, it is considered ‘Research Grade’. All observations are made available for download on iNaturalist and Research Grade observations are uploaded to the Global Biodiversity Information Facility (GBIF), depending on license requirements, where they are made available to scientists around the world. The DeLuca Bioblitz used iNaturalist as the central tool to collect and engage with participants of the bioblitz, where only observations from

iNaturalist were included in the bioblitz results. iNaturalist was chosen as the bioblitz platform because it is free, has many users, is user-friendly, and the resulting data are increasingly used in ecological research (Mason et al., 2025b). At check-in, participants were asked if they were familiar with iNaturalist, and those who were unfamiliar with the platform were given a brief tutorial on how to download and engage with the app.

Analysis of records

We obtained all iNaturalist data for the DeLuca Bioblitz efforts by using the iNaturalist data export tool on July 1, 2025. Data is from the four different bioblitz efforts, described above, hereafter referred to as the DeLuca Bioblitz. For each species with a Research Grade observation(s), we also collected the total number of observations of that species in (a) Osceola County, (b) the state of Florida, and (c) the globe using the iNaturalist API (iNaturalist 2025). We report the relative frequency of these species found as part of the bioblitz efforts (site-level) relative to the total number of observations of each species in Osceola County, as well as the state of Florida (regional-level). Additionally, we used the iNaturalist API to obtain species origin information in relation to Florida (i.e., introduced, endemic) for all species recorded during DeLuca Bioblitz. To assess the extent to which the bioblitz filled in locations in species distributions, for every species with fewer than 100 observations from the DeLuca Bioblitz, we found the next closest Research Grade observation outside of DeLuca Preserve. To do this, we used all Research Grade iNaturalist data in Florida through June 27th, 2025, downloaded using the iNaturalist data export tool. While using the Global Biodiversity Information Facility (GBIF) is the recommended approach for obtaining Research Grade data, we noticed that the taxonomy did not fully align between iNaturalist-exported data and GBIF. Therefore, we chose to use

iNaturalist data export, so our methods aligned with our methods for obtaining the DeLuca Preserve data.

We also compared the cumulative spatial sampling effort from the DeLuca Bioblitz to other randomly distributed locations across Florida. To do this, we used the create_square function in the sf package (Pebesma and Bivand 2023; Pebesma 2018) and randomly generated 100 different polygons (i.e., square boundaries) in the state of Florida that matched the size of the DeLuca Preserve (Figure S1). In each of these randomly generated polygons, we gathered the number of observers, number of observations, and number of species documented at Research Grade level using the iNaturalist API. Comparisons between the DeLuca Bioblitz and the randomly generated polygons did not have the same temporal resolution. The DeLuca Bioblitz cumulatively represented data collected on four days from four separate years, whereas the randomly generated polygons included all-time records (i.e., there was no temporal filter applied to the polygon-generated data).

Data analysis and availability

All data analysis was conducted in R (R Core Team 2025) and relied heavily on the Tidyverse workflow (Wickham et al. 2019). All data and code are available in a Github repository (https://github.com/coreytcallaghan/DeLuca_bioblitz_analysis) and will be archived in Zenodo.

Results

Overview of biodiversity outcomes

Across four annual bioblitzes, concentrated sampling at the DeLuca Preserve generated a dense, taxonomically diverse dataset (N=892 Research Grade species) that revealed numerous locally rare, novel, and under-documented species, while contributing disproportionately to regional biodiversity records. The highest reported species were pink sundew (*Drosera capillaris;* 210 observations), fourpetal St. Johnswort (*Hypericum tetrapetalum;* 207 observations), shiny blueberry (*Vaccinium mrysinites;* 159 observations), saw palmetto (*Serenoa repens;* 155 observations), and orange milkwort (*Senega lutea*; 149 observations). A total 45% of Research Grade species were documented two times or less, and 31% were only documented by a single observation. The spatial distribution of observations was heterogeneous across the preserve, with a pronounced concentration in areas of high accessibility (Figure 2). Species accumulation curves revealed a pronounced long-tailed distribution of species detection frequency and a non-asymptotic trajectory, indicating that species richness continues to increase with each sampling effort and that sampling completeness has not yet been achieved (Figure 3), suggesting that more species will be found with ongoing efforts.

Using iNaturalist data to define species origin, we also examined the accumulation of Florida-endemic and introduced species across the four bioblitzes (Figure S2). Both groups exhibited upward trajectories consistent with the overall species accumulation curve, indicating that continued sampling will likely reveal additional taxa. To date, 69 introduced and 20 endemic species have been recorded; however, these counts are provisional and likely underestimate true totals due to incomplete sampling and ongoing data curation.

Comparison with random landscape benchmarks

When comparing with randomly-sampled polygons, we found that the efforts associated with the DeLuca Bioblitz led to markedly higher observations per km² (118.0 vs. 20.1 ± 44.7; $t_{99} = -21.9$, $p < 0.001$) and species per km² (7.98 vs. 4.28 ± 3.77; $t_{99} = -9.80$, $p < 0.001$) than the statewide average, while observers per km² did not differ significantly (1.82 vs. 2.40 ± 3.79; $t_{99} = 1.52$, $p = 0.13$) (Figure 4). Shannon's diversity index further confirmed high diversity of observations at the preserve (H' = 5.81), approaching statewide diversity (H' = 6.82), indicating that the DeLuca Bioblitz captured both common and rare taxa in a relatively short time frame compared to the entire time frame for the 100 random polygons.

For species with less than 100 observations recorded during the DeLuca Bioblitz, we found that the average median linear nearest distance of the species recorded outside of the DeLuca Preserve was 7.043 km and ranged from 0.0035 to 7,595 km. The nearest observations represent common species such as white beggarticks (*Bidens alba*), ballmoss (*Tillandsia recurvata*), and Pickerelweed (*Pontederia cordata*). The species that was furthest away was false smut of palm (*Graphiola phoenicis*), a fungal leaf disease that affects palm species, where the nearest observation was in Lisbon, Portugal. Though there are no other Research Grade iNaturalist observations in Florida, this species has been known to occur in this region (Elliott 2005). Upon speaking with an expert in this field, they agree with the genus-level identification of this species, *Graphiola*, but did not agree that species-level identification is possible based on photographic evidence, given that there are multiple species of *Graphiola* worldwide (Piepenbring et al. 2012). Nevertheless, this is still the first observation of *Graphiola* on iNaturalist in the United States.

An analysis of relative rarity

Comparison with county- and state-level biodiversity baselines revealed that the DeLuca Bioblitz contributed disproportionately to records of regionally rare and infrequently documented species. A subset of species recorded at DeLuca (2 species: *Condylostylus flavipes* and *Aethes triassumenta*) were observed only within the preserve and had not been documented elsewhere in Florida (Figure 5). Bioblitz-derived observations accounted for a disproportionately high share of county-level occurrence records, particularly within Osceola County, where many species were represented primarily or exclusively by DeLuca Bioblitz observations (Figure S3).

Discussion

Bioblitzes are increasingly popular events (Figure S4), concomitant with a general increase in participatory citizen science in general (Feldman et al. 2021). While there are many benefits of such events for both engagement and natural resource management (Mason et al. 2025b), we provide a framework and an assortment of biodiversity metrics that can be used to evaluate the utility of bioblitzes for collecting relevant biodiversity data. Using a four-year case study, we show that a concentrated bioblitz effort can (1) generate dense, taxonomically diverse datasets over short time frames, (2) detect rare, endemic, and introduced species of management relevance, and (3) contribute disproportionately to regional biodiversity records. We evaluate potentially valuable data both for local, place-based adaptive management of natural resources, where, in this instance, observers found at least 33 endemic species and 71 introduced species. We also illustrate how data from a bioblitz can contribute to broad, regional-scale knowledge about biodiversity, with biodiversity metrics from the DeLuca Bioblitz matching or exceeding those from randomly selected polygons across Florida despite the much shorter sampling

duration. Overall, our results demonstrate that concentrated bioblitz efforts can generate essential biodiversity information for natural resource management.

The value of a given biodiversity observation is context-dependent. Biodiversity observations may be locally common species, new county occurrences, threatened species, or the first detection of a potential invader. However, we note that there are often context-dependent qualitative findings, where a single record of a specific species can be very valuable, such as a finding of a non-native species (Yeshwanth et al. 2025) or of a species of conservation concern (Wilson et al. 2020; Mesaglio et al. 2021). The DeLuca Bioblitz uncovered several notable findings, including a critically imperiled species, the Florida black dark bee (*Stelis ater*), along with multiple threatened (e.g., *Hemiargus ceraunus*) and endemic species (e.g., *Lupinus floridanus*), underscoring the conservation value of this property and the importance of continued biodiversity monitoring through the bioblitz. Since bioblitzes concentrate effort in space and time, they are especially powerful in this regard, acting as early-detection systems that complement ongoing monitoring programs. Further, when enough observations accumulate at a local site, they provide the foundation for mapping biodiversity patterns with increasing resolution (e.g., Figure 1). As analytical approaches for citizen science data continue to advance (Beninde et al. 2023; Stewart et al. 2023), such datasets could allow researchers not only to document presence but also to estimate relative abundance and detect temporal trends. In this way, repeated and well-organized bioblitzes could form the basis of long-term biodiversity monitoring at the site scale.

At a broader scale, we situated the DeLuca Bioblitz efforts in a statewide context by leveraging a random polygon approach to benchmark the biodiversity data generated from a short-temporal scale bioblitz. In terms of observations and species per unit area, the DeLuca Bioblitz efforts contributed 487% greater observations per km² and 86.5% greater species per km² than the mean of random polygons, despite the dramatically shorter time frame. Our random polygon analysis (Figure 4; Figure S1) provides a way to benchmark bioblitz contributions against broader iNaturalist activity. However, we acknowledge that underlying biodiversity, habitat composition, and accessibility vary across polygons, and observer behavior on iNaturalist is not spatially random, all of which can contribute to variation among polygons. As such, these comparisons are best interpreted as coarse contextual benchmarks rather than direct ecological contrasts. Nevertheless, this analysis illustrates that bioblitzes can produce unusually dense datasets, creating snapshots of biodiversity that provide a complementary and powerful perspective. We further conducted an exploratory analysis, comparing DeLuca Bioblitz data to iNaturalist data from 39 Florida state parks (Lowe et al., 2026), to evaluate active versus passive sampling (Figure S5). Despite the DeLuca Bioblitz data being collected over just four days, compared to the full temporal extent of the state park data, the bioblitz yielded more observations and species than expected based on state park trends. This illustrates the value of bioblitzes in generating more observations and species than passive sampling. Given that bioblitzes can generate robust species inventories in 24 hours that rival entire temporal scales in our random polygons, land managers should consider the benefits of these non-traditional sampling events to create baseline species inventories. Lastly, conservation practitioners interested in both rare and potentially invasive species can leverage bioblitz events to document large amounts of data rapidly, compared to more traditional monitoring methods that can be more costly and time-consuming.

Our evaluation focused on quantitative metrics that are broadly applicable across bioblitz events, but this represents only one dimension of their value. Future work could strengthen these insights by directly comparing bioblitz outcomes with professionally collected survey data, assessing how well volunteer-collected records align with standardized monitoring (Katzer et al. 2025). Additional approaches might include examining taxonomic biases (e.g., toward charismatic or easily photographed taxa or taxonomic uncertainty), quantifying participant learning or engagement outcomes (Echeverria et al. 2021; Niemiller et al. 2021), or integrating bioblitz data with other monitoring tools such as eDNA or remote sensing (Granqvist et al. 2025; Cavender-Bares et al. 2022). Importantly, the results we present here emphasize short-term outcomes—the concentrated biodiversity snapshots generated over the course of a single event. The longer-term value, particularly the potential for repeated bioblitzes to build cumulative datasets that enable site-level monitoring and adaptive management, remains to be fully realized. As such, bioblitzes represent not only a means to engage the public but also an underutilized mechanism for generating robust, repeatable biodiversity data with clear relevance for conservation and management.

Conclusions

We provide an overview of how a bioblitz can provide valuable biodiversity data. While we introduce quantitative metrics and provide a conceptual framework (Figure 1) for how data from a bioblitz can contribute to both local place-based natural resource management as well as regional-scale biodiversity research. However, we stress that hosting a bioblitz is time-consuming and requires logistical planning (Tiago et al. 2024). Nevertheless, based on our

results and collective experience, we suggest that natural resource managers should consider the use of bioblitzes and how they can contribute to adaptive management goals.

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Figures and Tables

Table 1. Metrics used to assess bioblitz biodiversity utility. Metric scale is defined as site-level (SL) which characterizes biodiversity patterns within the site defined for the bioblitz and regional-level (RL) which contextualizes bioblitz observations relative to broader regional biodiversity data. We additionally provide a brief definition of each metric and the methodology used to calculate them.

Metric	Scale	Definition	Calculation
Total observations	SL	Total observations taken at the site.	Count of Research Grade observations.
	RL	The relative number of observations at the site compared to the surrounding region.	One hundred random polygons that were the size of the site were generated within the region. For each polygon, the number of observations was calculated and compared to the corresponding values from the site.
Number of observers	SL	Unique observers who submitted observations at the site.	Count of unique observers from Research Grade observations.
	RL	The relative number of observers at the site compared to the surrounding region.	Follows same methodology as total observations (RL) but compares observers.
Number of species	SL	Number of unique species observed at the site.	Count of unique species from Research Grade observations.
	RL	The relative number of unique species at the site compared to the surrounding region.	Follows same methodology as total observations (RL) but compares number of unique species.
New species	SL	Measure of new species at the site based on sampling effort.	Species accumulation curve.
Relative frequency	RL	Frequency of species at the site relative to the region.	For each species, proportion of observations at the site is divided by its proportion of observations in the region.
Relative rarity	RL	Measure of species rarity based on number of observations at the site and region.	Relative frequency is used to define categories for rarity, including site-only, locally rare, common everywhere, rare everywhere, and underreported everywhere.
Relative diversity of species	RL	Diversity of species at the site compared to the region.	Shannon's diversity index.
Nearest external observation	RL	Distance from the site to the nearest external observation for species with fewer than 100 site-level observations.	For each qualifying species, the minimum distance to any observation outside the site boundary was computed.

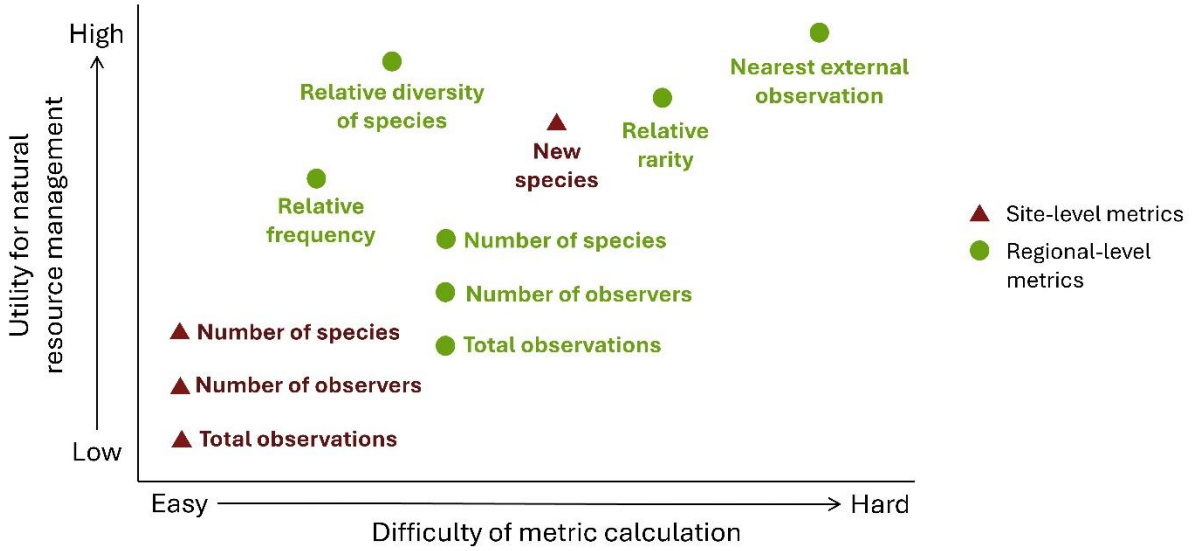

Figure 1. Conceptual diagram showing the utility and difficulty of calculating the bioblitz biodiversity utility metrics explored in this paper. Metrics are grouped as either site-level (defined relative to the bioblitz site) or regional-level (defined as the contribution of the bioblitz site to the broader region). See Table 1 for definition of each metric.

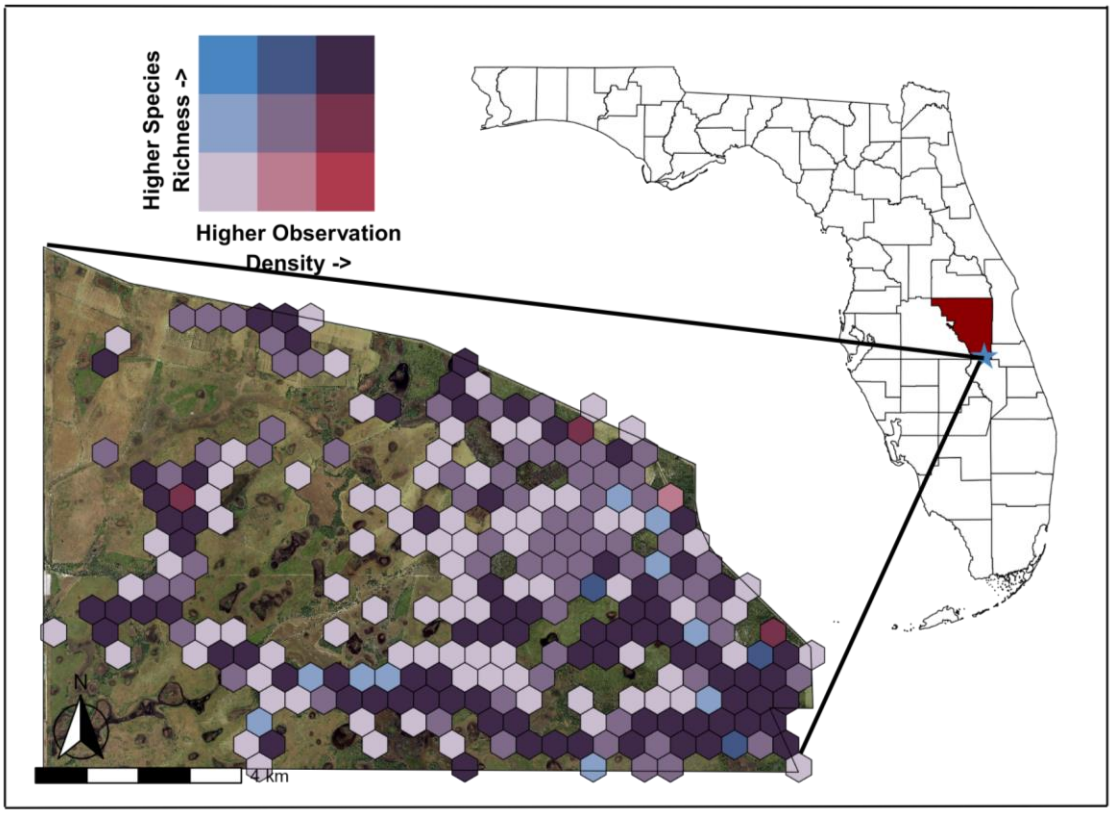

Figure 2. A map of DeLuca Preserve, showing its location in Florida and its relative remoteness. Osceola County, where the preserve is located, is highlighted in red. Additionally displayed is a density heat map of where the observations within the preserve have been derived from the DeLuca Bioblitzes, and the concentration of observations and species richness. Bivariate colors reflect quantile-based classification (dim = 3) of observation density and species richness.

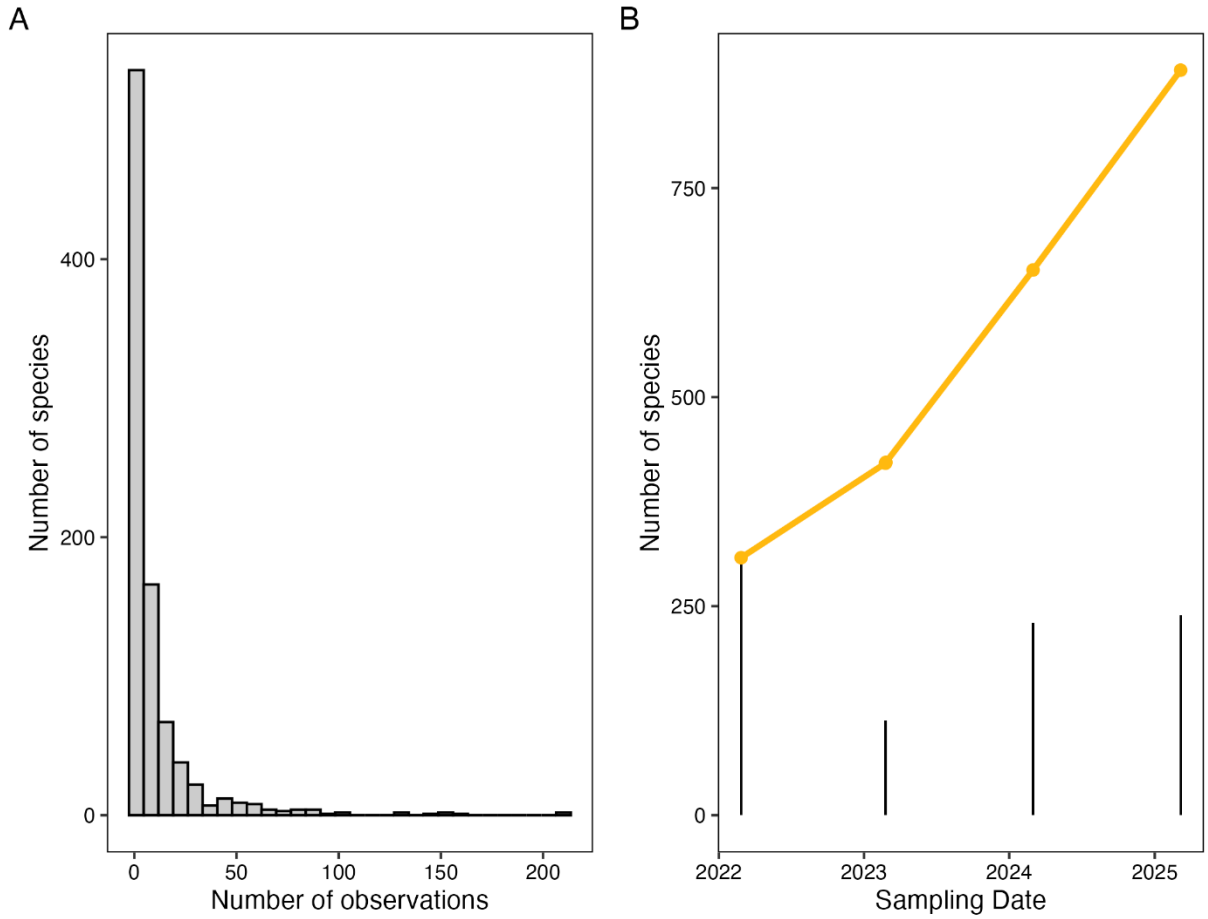

Figure 3. (A) Frequency distribution of species detection rates across the DeLuca Bioblitz dataset, showing the number of species in each observation group. (B) Species accumulation across sequential bioblitz events (2022–2025), indicating continued discovery of novel taxa and incomplete sampling saturation. The vertical lines represent the number of new Research Grade species found that year.

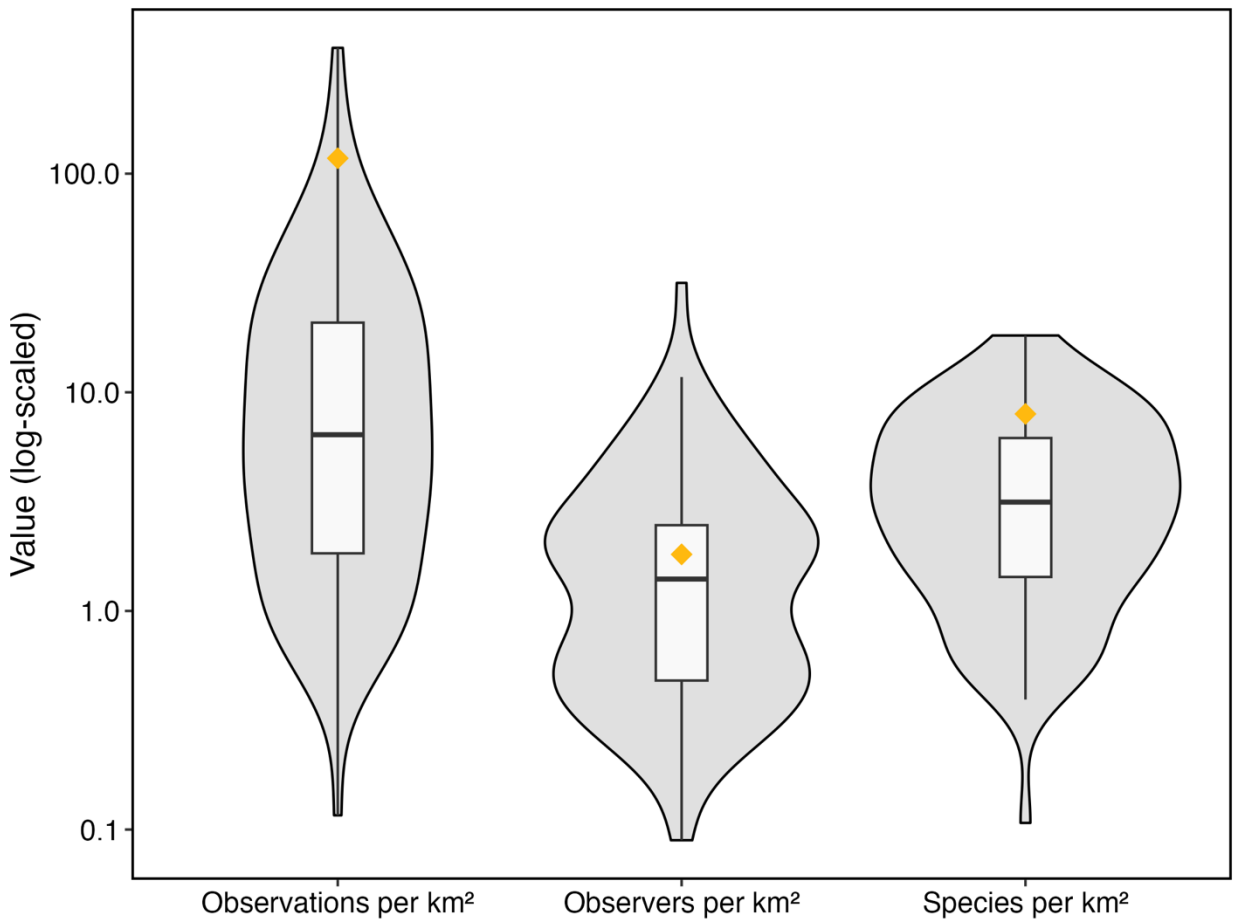

Figure 4. Comparison of biodiversity metrics at the DeLuca Preserve relative to 100 randomly selected polygons of equivalent area across Florida. Violin and boxplots represent the statewide distribution of iNaturalist data in random polygons for three metrics normalized by area (observers, observations, and species per km²), while the diamond markers indicate values from the DeLuca Bioblitz dataset. DeLuca exhibited markedly higher observation density (118.0 vs. 20.1 ± 44.7; $t_{99} = -21.9$, $p < 0.001$) and species density (7.98 vs. 4.28 ± 3.77; $t_{99} = -9.80$, $p < 0.001$) than the statewide average, whereas observer density did not differ significantly (1.82 vs. 2.40 ± 3.79; $t_{99} = 1.52$, $p = 0.13$).

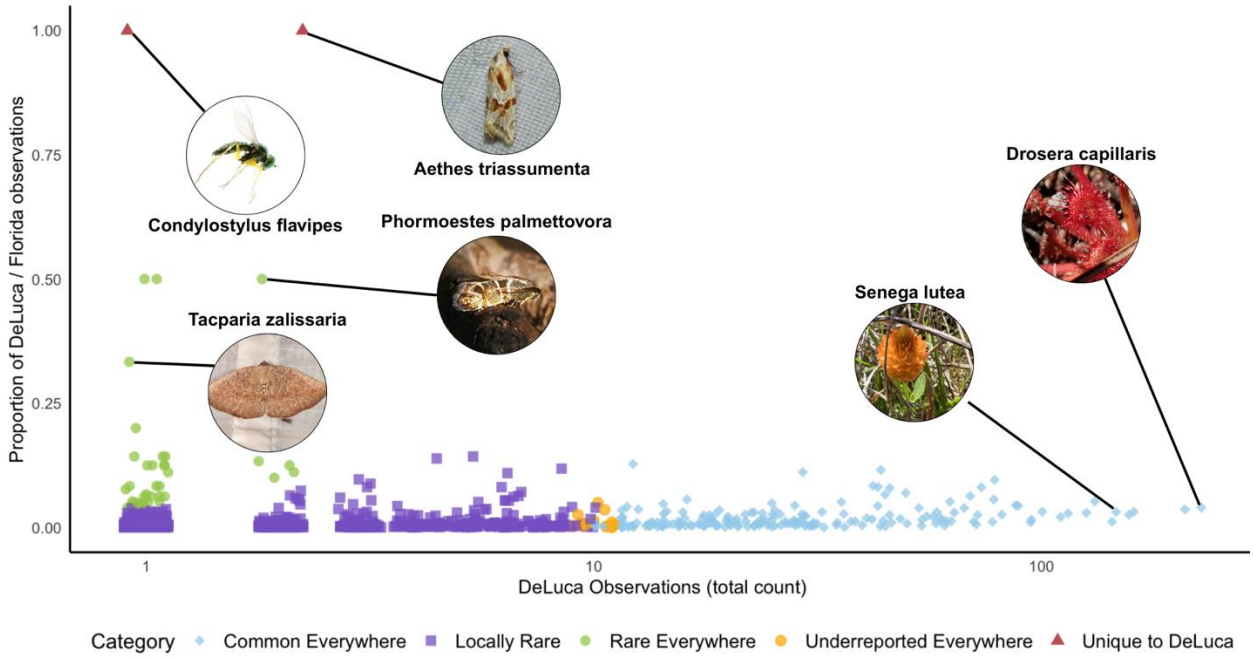

Figure 5. Each point represents a species observed at DeLuca, plotted by its total observation count at DeLuca (log-scale) versus its proportional representation relative to Florida-wide observations. Colors indicate rarity categories: blue represents common everywhere (> 10 observations in DeLuca; > 25 observations in Florida), yellow represents underreported everywhere (= 10 observations in DeLuca; <= 25 observations in Florida), purple represents locally rare (< 10 observations in DeLuca; > 25 observations in Florida), green represents rare everywhere (< 10 observations in DeLuca; < 25 observations in Florida); and red represents unique to DeLuca. Image attributions: Odin Toness (*Condylostylus flavipes*), Laura Gaudette (*Tacparia zalissaria*), Jeff Niznik (*Aethes triassumenta*), Liz Childress (*Phormoestes palmettovora)*, Aurora Alcaide (*Senega lutea*), and Donald Fraser (*Drosera capillaris*).

Supplemental Material
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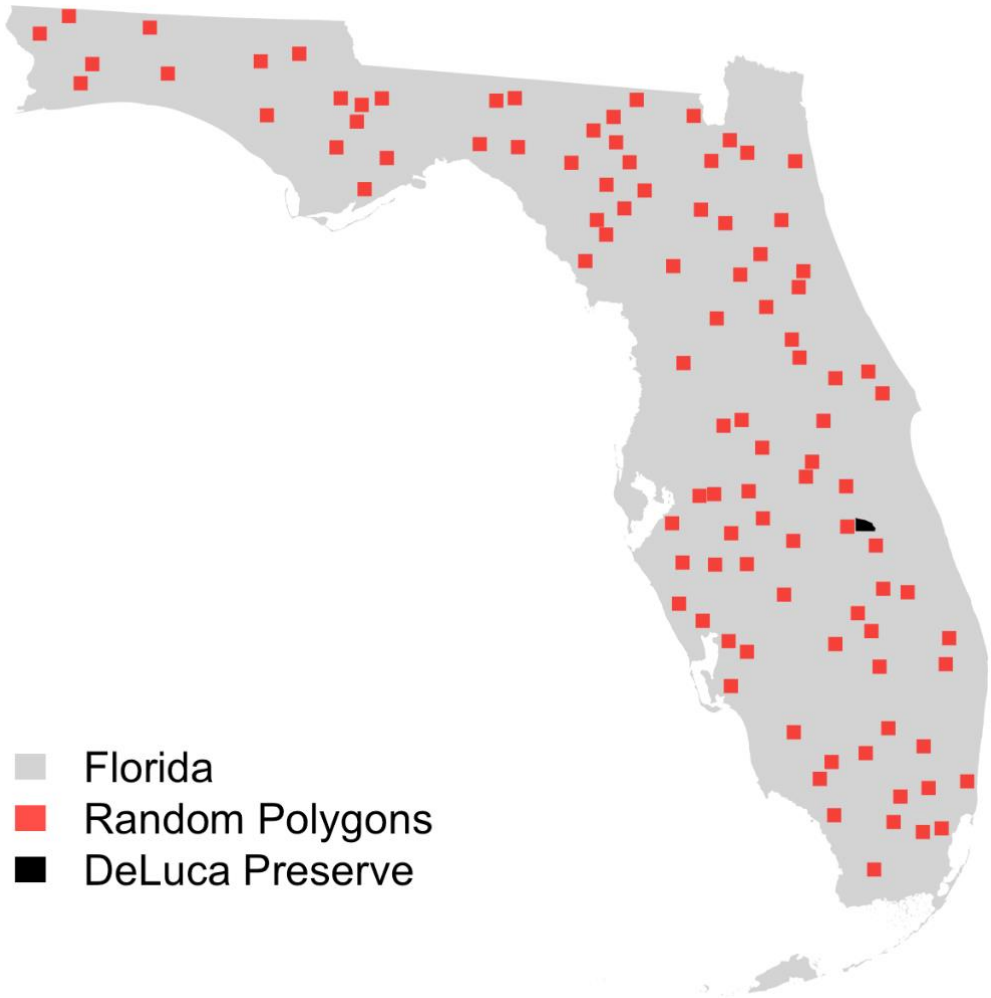

Figure S1. To determine the cumulative effort and illustrate the potential people-hours compared to other locations in Florida, we generated 100 random square boundaries with the same area as the DeLuca preserve across Florida.

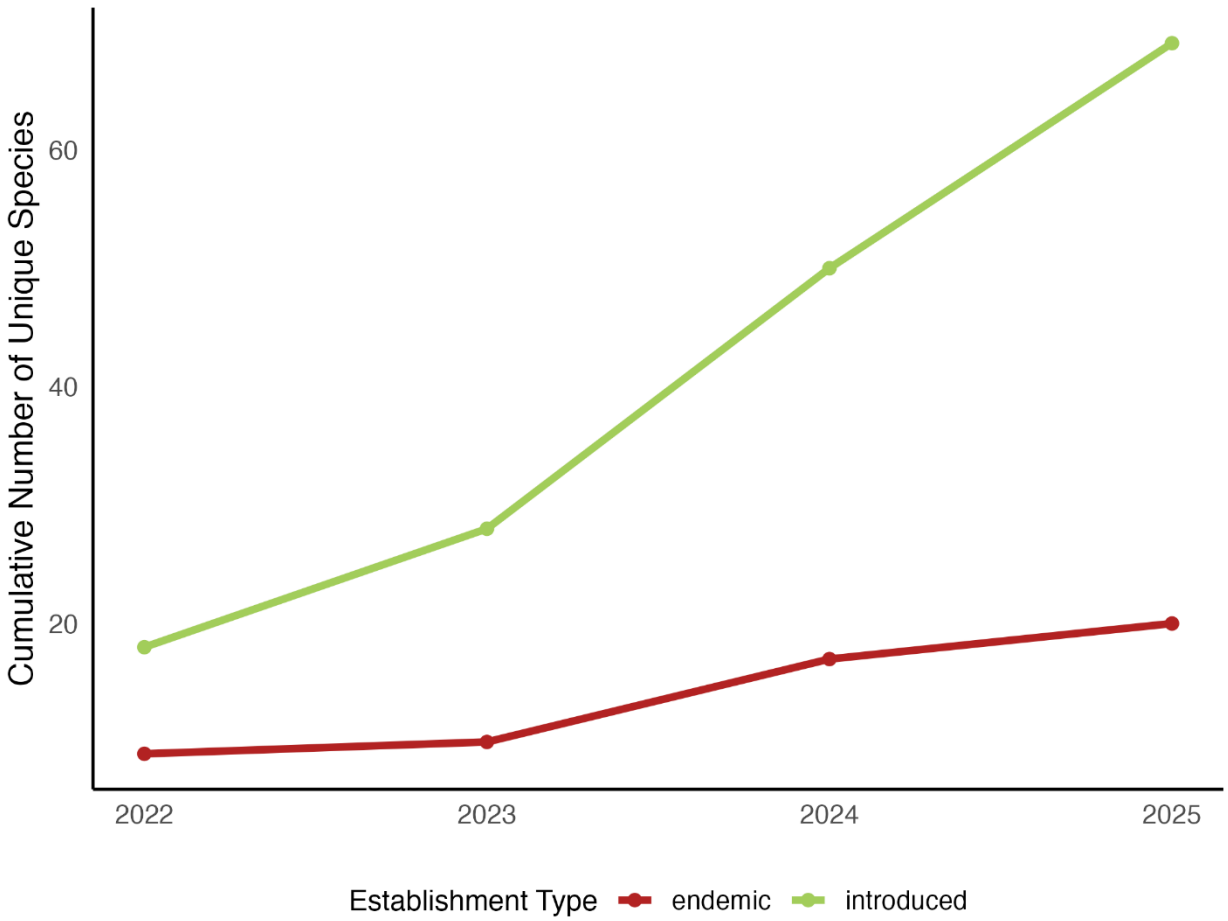

Figure S2. Cumulative number of endemic and introduced species recorded during bioblitz events at DeLuca Preserve, based on Research Grade iNaturalist observations. The orange line represents the number of introduced species over time. The blue line represents the number of endemic species to Florida over time. Both represent an increase in findings of establishment type per sampling effort.

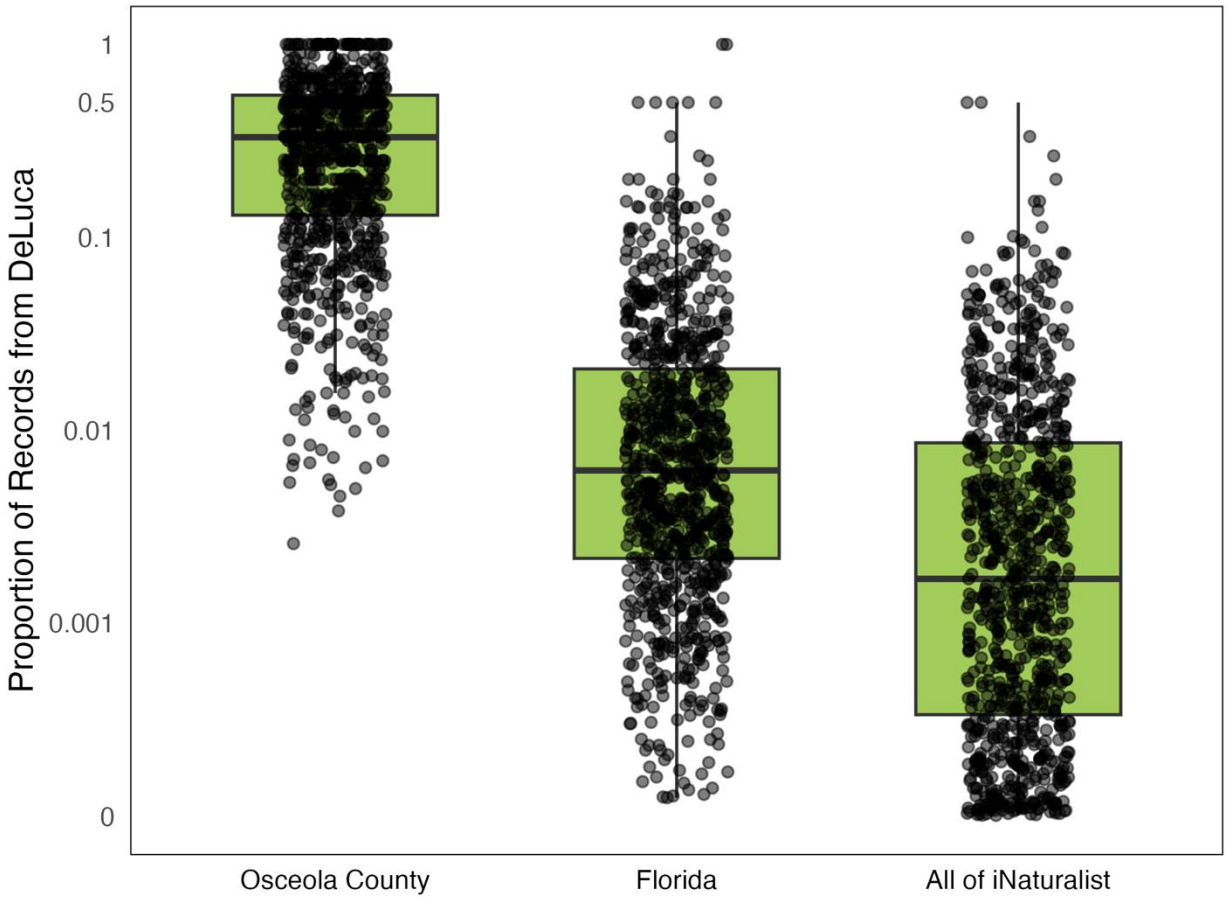

Figure S3. Proportion of all recorded observations per species that originated from the DeLuca Bioblitzes across different spatial scales. Each point represents a single species, with boxplots denoting the interquartile range. The three panels correspond to (from left to right): observations within Osceola County, all observations across Florida, and all observations on iNaturalist. All observations are Research Grade.

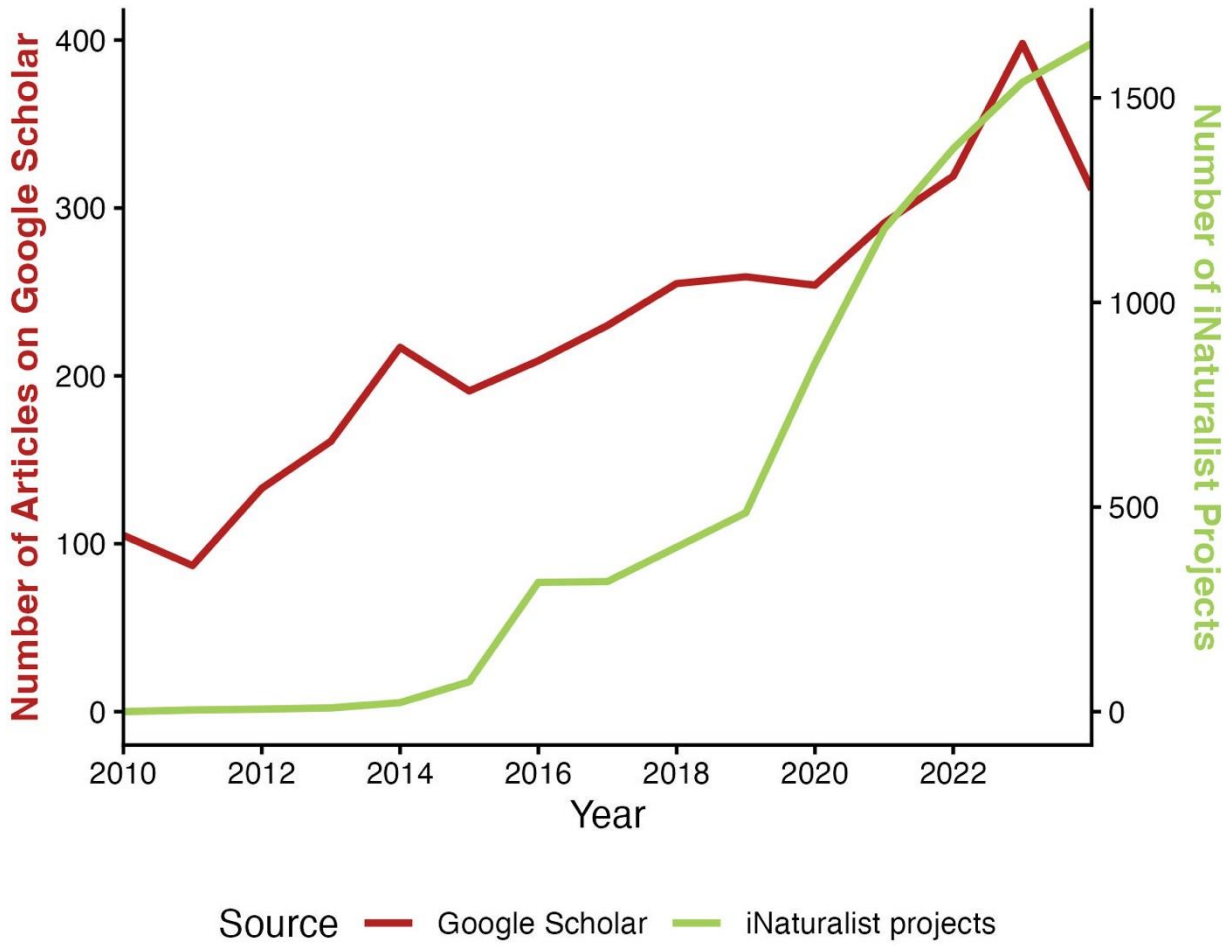

Figure S4. The blue line represents the number of articles returned from Google Scholar searches for the term "Bioblitz" by year. The orange line represents the number of iNaturalist projects that contain the term "Bioblitz" by year. Both cases illustrate the increasing popularity of bioblitzes over time.

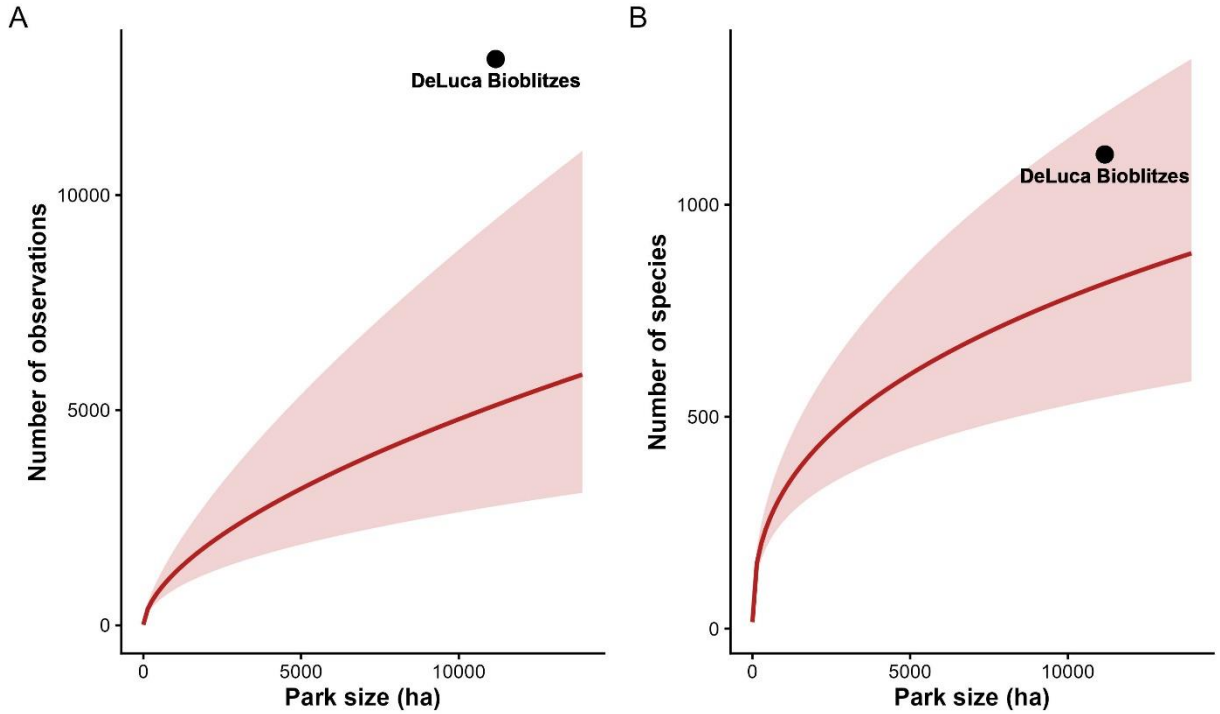

Figure S5. To explore how DeLuca Preserve Bioblitz observations compare to observations in Florida state parks, we modeled the relationship between number of observations and number of species across 39 state parks (see Lowe et al., 2026). We ran two negative binomial generalized linear models: one with the number of observations as the response and log-transformed park size (ha) as the predictor, and one with the number of species as the response and log-transformed park size (ha) as the predictor. Model structure was chosen based on residual diagnostics using the *DHARMa* package in R (Hartig, 2024). We then plotted the fitted trend lines with standard error, and highlighted data from the DeLuca Bioblitzes with a point showing where it falls on each graph. Note that the state park data covers all dates in iNaturalist, whereas DeLuca Preserve Bioblitz represents only observations taken during four bioblitzes. This plot illustrates the value of bioblitzes to collect more observations and species than passive observation.

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January 28th, 2026

Dr. Tara Cornelisse
Lead Editor
Natural Areas Journal

Dear Editor,

On behalf of my co-authors and myself, I would like to formally resubmit our manuscript titled "*Bioblitzes provides valuable biodiversity data*" for publication in the *Natural Areas Journal*.

We thank you for your consideration of the manuscript and the constructive reviews which helped us significantly improve this paper. We have listed all the referees' comments below and describe how we have responded to each suggestion. During the resubmission we uploaded a 'clean' version and a 'tracked changes' version to clearly highlight the changes made to our manuscript.

We hope that the revisions to the manuscript meet with your satisfaction, making this a valuable contribution to the *Natural Areas Journal*.

Sincerely,

Corey T. Callaghan (he/him)
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Department of Wildlife Ecology and Conservation
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UF UNIVERSITY of FLORIDA

Editorial Comments

The reviewers found your paper to be a valuable contribution to the literature, however there are some revisions requested. Reviewer #2 found the paper to be fairly well-written, while Reviewer #3 found it to be somewhat unorganized and repetitive. I recommend carefully reviewing the manuscript to remove instances of repetition and ensuring that any colloquial language is replaced with more scientific terminology. Please address the more specific comments of both reviewers as well, and make sure all figures are accessible, important, and well-explained. The comments from both reviewers are below; I look forward to receiving your revised manuscript.

Response: *We appreciate this thorough and positive set of reviews from the Reviewers. We have addressed each comment below, point by point, but highlight here the following overarching edits:*

- *We have revised the narrative so that it is clear it is (a) a framework and then (b) a case study using that framework. This included adding an additional paragraph at the end of the introduction focused on the framework itself.*
- *We edited the manuscript to remove repetition and minimize colloquial language throughout as suggested by Reviewer 2.*
- *We completed an additional exploratory analysis as suggested by Reviewer 1, comparing the results to more active samplin (e.g., at State Parks).*

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Reviewer 2 Comments

This paper is a welcome addition to the literature. We have utilized bioblitzes in the past, and it is good to see that they are an excellent way at generating biodiversity data. I don't make this a requirement, but it would be interesting to compare the data you got to state parks or other protected lands that get regular visitors - so passive data collection vs active data collection. You can also look by taxon. These ideas may be beyond the scope of the paper, and I think the paper adequately demonstrates that bioblitzes do work.

Response: *We thank the Reviewer for these positive comments about our manuscript, and likewise hope it is a useful addition to the literature. Your idea is a good one! We actually had a recent paper which investigated the data from parks in Florida, and compared citizen science with state agency data (here: https://doi.org/10.1002/2688-8319.70173). We think this is a good idea by the Reviewer and as such we performed an exploratory analysis which used the parks from this paper (N=39) and put them in the context of the bioblitz (active sampling). We made an additional supplemental figure and added a couple lines about this to the discussion (Lines 284-289):*

> *"We further conducted an exploratory analysis, comparing DeLuca Bioblitz data to iNaturalist data from 39 Florida state parks (Lowe et al., 2026), to evaluate active versus passive sampling (Figure S5). Despite the DeLuca Bioblitz data being collected over just four days, compared to the full temporal extent of the state park data, the bioblitz yielded more observations and species than expected based on state park trends.".*

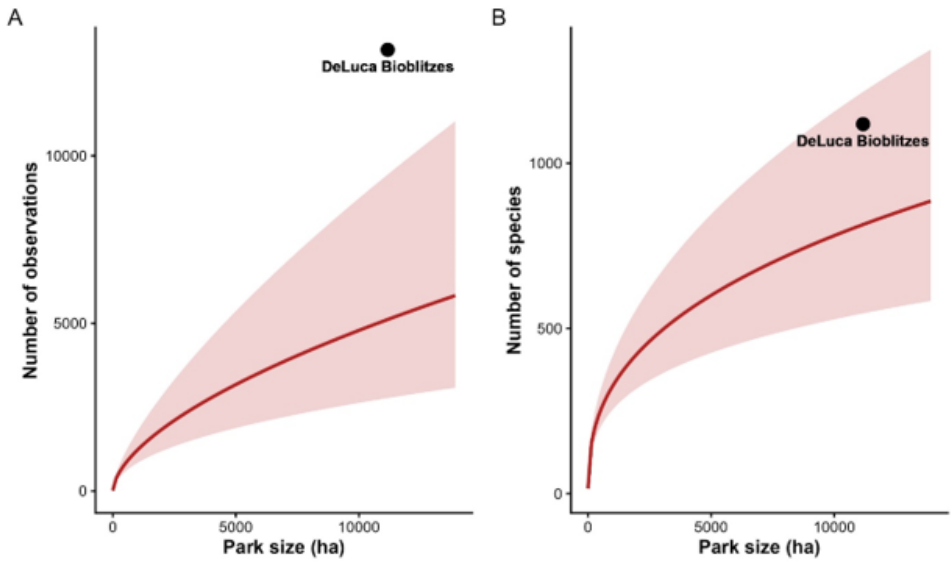

Figure S5. To explore how DeLuca Preserve Bioblitz observations compare to observations in Florida state parks, we modeled the relationship between number of observations and number of species across 39 state parks (see Lowe et al., 2026). We ran two negative binomial generalized linear models: one with the number of observations as the response and log-transformed park size (ha) as the predictor, and one with the number of species as the response and log-transformed park size (ha) as the predictor. Model structure was chosen based on residual diagnostics using the *DHARMa* package in R (Hartig, 2024). We then plotted the fitted trend lines with standard error, and highlighted data from the DeLuca Bioblitzes with a point showing where it falls on each graph. Note that the state park data covers all dates in iNaturalist, whereas DeLuca Preserve Bioblitz represents only observations taken during four bioblitzes. This plot illustrates the value of bioblitzes to collect more observations and species than passive observation.

I just have a few minor comments.

83. This is far too great of precision for the coordinates. 6 decimals is 43-111 mm, which is way too precise for a land area like this. Switching to 4 decimal places barely moved the marker. 4 is sufficient and 5 is the maximum.

Response: *We agree - we switched to 2 decimal places which is more than sufficient.*

163 - spell out numbers under 10 (here are two examples of 4). See also 181 "1 or 2 times". There are a few other examples as well.

Response: *Suggestion adopted.*

You have some accessibility issues with the figures. To check whether colors can be distinguished, use a color blindness simulator like Coblis: https://www.color-blindness.com/coblis-color-blindness-simulator/

Response: *Thank you! We have revised the figure color schemes accordingly.*

Fig. 1. You can't tell the difference between red and purple under monochromacy. Although rare as a disorder, many people print papers out in black and white. Consider using shape in addition to color.

Response: *Good suggestion – we have added shape as well to the figure.*

Fig. 5 Many of the colors blend together under different forms of color blindness. Use shape in addition to color.

Response: *Suggestion adopted.*

Figure S2 might not be accessible. The orange and blue are almost indistinguishable under monochromacy. If you keep the same colors, I suggest that you move the labels to the actual lines on the graph instead of in a legend below. You should also not refer to colors in the legend. Same will go for Fig. S4.

Response: *We have gone through and checked these all and edited accordingly. Thank you for this suggestion.*

Reviewer 3 Comments

Hoskins et al. make a case for the value of bioblitzes for science, conservation and community engagement. They describe their efforts in the DeLuca Preserve in Osceola County, Florida, over 4 years. This kind of project is great, further recognition of citizen science data and documentation of outcomes from bioblitzes like mentioned here is warranted.

Response: *We thank the reviewer for their positive assessment of the manuscript and for recognizing the value of bioblitzes for science, conservation, and community engagement. We agree that documenting the scientific outcomes of bioblitz efforts is important, and we hope this case study helps further highlight the broader utility of citizen science data for biodiversity research and management. We also are grateful for their comments below, which helped improved our manuscript.*

The ms. is not very well written or organized. The writing is highly repetitive, using up space where more interesting info could go. The writing style is a bit wordy, folksy in places and some passages stray from merely awkward into inaccuracy. With this many authors, someone should have taken the time to edit the submission.

Response: *We thank the reviewer for this feedback and have taken steps to improve clarity, and organization in several parts of the manuscript. Specifically, we have edited the Abstract, portions of the Introduction, and the opening sections of the Discussion to reduce redundancy, and improve the overall flow. We believe these targeted revisions improve readability and flow without altering the scope or conclusions of the study. We will highlight a select few edits here. We refer the Reviewer to see the revised manuscript uploaded in our revision for all changes.*

- *We trimmed the abstract, to remove the restatement of the value of a bioblitz.*
- *We removed the descriptive text about the user numbers which appeared in the methods and results. It now only appears in the results.*
- *We have added more interesting information (i.e., more background on the conceptual framework)*
- *We removed repetitive framing language in the Introduction that restated the general value of bioblitzes and citizen science across multiple paragraphs, tightening the narrative to focus more directly on the specific research gap addressed by this study.*
- *We also reworked the table describing/introducing the metrics to be stand alone, and separate from Figure 1.*
- *We removed redundant language, for example now only refer to regional scale as opposed to regional and landscape scale previously meaning the same thing.*
- *We revised the opening of the Discussion to explicitly enumerate the key findings of the study rather than reintroducing background material, reducing redundancy between the Introduction, Results, and Discussion.*
- *We edited parts of the Discussion to focus on flow, and allow some of the main conclusions to come at the end.*

The bioblitz outcomes are great and more should be done to highlight key findings. The "Overall contributions summary" section at top of results is a weak or off-track summary of their efforts. Slightly uncommon metrics pop up elsewhere in ms also.

Response: *We thank the reviewer for highlighting the strength of the bioblitz outcomes and for this constructive suggestion. In response, we revised the Results and Discussion to more clearly foreground the key findings of the study, including the detection of rare, endemic, and introduced species and the disproportionate contribution of bioblitz data to regional biodiversity benchmarks. We added the following to the Discussion first paragraph to better highlight the key findings (Lines 242-245):*

> *"Using a four-year case study, we show that a concentrated bioblitz effort can (1) generate dense, taxonomically diverse datasets over short time frames, (2) detect rare, endemic, and introduced species of management relevance, and (3) contribute disproportionately to regional biodiversity records.".*

We also revised the opening Results subsection to emphasize biodiversity outcomes rather than event-level descriptors and renamed the section "Overview of biodiversity outcomes". We removed the event description text, as this was already stated in the results. Ultimately, we feel this helps the narrative and keeps it more on track throughout.

We understand the comment about slightly uncommon metrics. We have now thought critically about the narrative and made some edits, which helps alleviate this concern. See the comment below for details.

I like the comparison to numerous random polygons in the state, not sure why the section is called "Gap Filling". The idea is not as novel as they suggest. Additional discussion of the obvious criticisms of this approach would strengthen their conclusions. E.g. do underlying biodiversity, observer access, quality or motivation play any role? Any drawbacks to comparing the average of random polygons?

Response: *We thank the reviewer for this helpful feedback. In response, we renamed the section formerly titled "Gap Filling" to "Comparison with random landscape benchmarks" to more accurately reflect the purpose of this analysis as a contextual comparison rather than a claim of novelty. We also revised the text to avoid overstating novelty and to clarify that the random polygon approach is intended as a benchmarking exercise to place bioblitz-derived biodiversity data into a broader landscape context. In addition, we expanded the Discussion to explicitly acknowledge limitations, which reads (Lines 279-282):*

> *"However, we acknowledge that underlying biodiversity, habitat composition, and accessibility vary across polygons, and observer behavior on iNaturalist is not spatially random, all of which can contribute to variation among polygons. As such, these comparisons are best interpreted as coarse contextual benchmarks rather than direct ecological contrasts.".*

I felt a bit of a mismatch between the narrative and figures. The biodiversity metrics in Fig 1. perhaps should be better explained or integrated into text. Some figure legends are not self-explanatory.

Response: *This comment made us think critically about the narrative. Our main fix here to focus the narrative was to allow the 'framework' and 'analysis' to be complementary and not at odds with one another. We first made this clear by editing the last paragraph of the introduction (Lines 65-70):*

> *"First, we provide a replicable framework that can be expanded outside of our study site, with the aim of helping natural resource managers leverage bioblitzes to advance biodiversity data collection. We then use this framework to illustrate the scientific value of a bioblitz as a mechanism for generating biodiversity data, including baseline species inventories, novel species records, and indicators of regional biodiversity utility (Figure 1).".*

We then added an additional paragraph to the end of the introduction, following the 'goals' of the paper, which solely introduces the framework (Lines 74-87):

> *"Bioblitz biodiversity utility framework*
> *We present a conceptual framework for evaluating the biodiversity value of bioblitzes by organizing metrics along two dimensions: their biodiversity utility for management and monitoring, and the analytical complexity required to calculate them (see Figure 1). The framework distinguishes site-level metrics that characterize biodiversity patterns within the site defined for the bioblitz (this includes, total observations, number of species, number of observers, and new species) from regional-level metrics that contextualize bioblitz observations relative to broader regional biodiversity data (this includes, relative frequency, relative rarity, and relative diversity of species, sampling intensity, and nearest external observation). Together, these metrics capture complementary dimensions of biodiversity value, ranging from baseline inventory completeness to the contribution of short-term, concentrated effort to regional biodiversity knowledge. Because we separate site-level and regional-level metrics, this framework provides a flexible tool that natural resource managers can adapt to evaluate the biodiversity outcomes of bioblitzes across different regions, taxa, and management objectives.".*

This allows the reader to then be prepared for the biodiversity metrics discussed in more detail in the methods, and subsequently, the results. This explicitly addresses the previous comment, about uncommon metrics, by helping the flow be more natural, aligning the narrative and figures, especially Figure 1, which sets up the narrative for the rest of the paper. In addition, we split Figure 1 into a figure and a more comprehensive Table which introduces the metrics, utlimately improving the flow.

The key result of 1252 species (line 113) in 11,000 hectares is great, perhaps a bit low for a

preserve of that size, in that area, I suspect they will add many, many more. Line 174 mentions 892 (research grade) and it looks like it is up to ~1800 in Figure 3?

Response: *There was a mistake in Figure 3 which has now been fixed. We have now focused on the Research Grade total, as this makes the most conceptual sense to focus on. We also agree that more will be added, and revised the text accordingly (Lines 193-195):*

> *"with each sampling effort and that sampling completeness has not yet been achieved (Figure 3), suggesting that more species will be found with ongoing efforts.".*

Potential typos / minor suggestions:

line 122 - perhaps more than 200 million?

Response: *Yes – good catch! We have fixed this. Thank you.*

line 125 - any observations can be downloaded, not only research grade

Response: *Yes, true. We have edited this sentence to be more precise (Lines 132-135):*

> *"All observations are made available for download on iNaturalist and Research Grade observations are uploaded to the Global Biodiversity Information Facility (GBIF), depending on license requirements, where they are made available to scientists around the world.".*

line 157 - mention R as well

Response: *Good suggestion. We added the following section shortly after this placement to not break up the flow too much (Lines 176-177):*

> *"All data analysis was conducted in R (R Core Team 2025) and relied heavily on the Tidyverse workflow (Wickham et al. 2019).".*

line 169 - github link broken or still private

Response: *We have made it public! And will archive the repository on Zenodo once accepted.*

Tracked Changes Word Version

Click here to access/download;Supplemental Material;NAJ-S-25-00037.R1_tracked changes manuscript.docx

~~A bioblitz~~Bioblitzes provides valuable biodiversity data

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Abstract

Bioblitzes are increasingly popular participatory science events aimed at documenting as many species as possible over a pre-defined period. They enhance community engagement and increase scientific literacy, while also contributing ~~valuable biodiversity data. Bioblitzes can generate~~ immense amounts of biodiversity data in a relatively short time span. ~~But~~Yet, the value of ~~such intense~~the biodiversity data ~~are~~generated is rarely quantified. In this study, we ~~use~~used the DeLuca Bioblitz as a case study to develop a biodiversity utility framework and a list of biodiversity metrics to evaluate the biodiversity data contributions from bioblitzes. We ~~calculate~~calculated summary statistics ~~from iNaturalist data collected during the bioblitz~~, including the total number of observations, number of observers, and number of species from the DeLuca Bioblitz. ~~Then we compare~~ We compared these ~~data~~summary statistics to random polygons of the same area across ~~the region~~Florida, to evaluate the impacts of bioblitz efforts compared to participatory science generally. ~~We found that bioblitzes~~Bioblitzes tend to generate more biodiversity data when compared to other random polygons of the same size throughout the state of Florida~~, especially when considering the temporal comparison. Our results illustrate the potential of bioblitzes to generate valuable biodiversity data.~~. Importantly, Bioblitz data was collected over a 24-hour period for 4 years, yet it outpaced biodiversity contributions in random polygons with data collected over multiple years on iNaturalist. The bioblitz in our case study also generated key records of new invasive species, new continental records, and records of imperiled species in Florida. Together, these data can help land managers and conservation professionals collect robust data, over a short duration, to create baseline species inventories and document species of concern for both management and conservation.

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Keywords: community science; citizen science; adaptive management; bioblitz; iNaturalist

Introduction

Site-level species inventories, such as a list of possible species in a certain property or park, often serve as baseline data for natural resource managers (Maes et al. 2015; Katzer et al. 2025). This information is essential for natural resource managers to prepare management plans and monitor ecological changes over time, particularly as species shift their ranges (Weiskopf et al. 2020) and local extinction and colonization events become increasingly common (Cowie et al. 2022; Storch et al. 2022). However, gathering such data can be time-consuming and costly; therefore, inventories can be difficult to maintain (Katzer et al. 2025). Advances in technology have given rise to many long-term monitoring platforms, including participatory citizen science, sometimes referred to as community science or citizen science. Participatory citizen science is increasingly contributing to biodiversity research (Fauzia et al. 2023; Mason et al. 2025a), conservation (Hsu et al. 2025), and natural resource management (McKinley et al. 2017; Mason et al. ~~2025b).~~2025b), for example by quantifying species distributions (Zulian et al. 2021) or detecting biological invasions (Larson et al. 2020). The bulk of all biodiversity data in the world now comes from platforms such as eBird and iNaturalist—upwards of 80% of all global biodiversity data since 2010 (Callaghan et al. 2020). ~~Citizen science data are increasingly used to answer questions about species distributions (Zulian et al. 2021), to monitor biodiversity in a given geographic region (Tiralongo et al., 2020), and to detect biological invasions (Larson et al. 2020).~~

Bioblitzes are one mechanism in which participants are encouraged to contribute data through participatory science. A bioblitz has many definitions (Meeus et al. 2023), but in general, it involves a set time period where the goal is to document as many species as possible within a

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pre-defined geographic region (Meeus et al. 2023). This can range from a ~~given~~single park~~, or~~ to large geographic regions. One particularly well-known example is the City Nature Challenge — a four-day bioblitz organized annually by the California Academy of Sciences — where cities across the world participate in a friendly competition to get the most observations, species, and people using iNaturalist (Palma et al. 2024; Kishimoto & Kobori 2021; Sakurai et al. 2022). Bioblitzes are useful for engaging the public in conservation and creating a sense of personal involvement in environmental research (Roger & Klistorner 2016). They are also useful for generating large amounts of data in a given location (Meeus et al. 2021, 2023). The City Nature Challenge alone provides the most data of any week to the iNaturalist platform each year (Meeus et al. 2023; Palma et al. 2024). ~~However, the majority of~~Despite their global popularity, most bioblitzes occur at localized scales, ~~where potential participants are readily engaged (Young et al. 2021). Yet~~and consequently their scientific value ~~is~~has been less ~~often emphasized, particularly in terms of the biodiversity data they can generate for a specific site~~frequently quantified.

Bioblitzes can have a variety of outcomes, ~~both for~~ including building community, increasing engagement in science, incentivizing ~~people to continue using~~continued participation on platforms like iNaturalist, and ~~also for their~~generating biodiversity data (Meeus et al. 2021, 2023~~). Bioblitzes build community, increase engagement in science, and serve as educational tools (~~; Pollock et al. 2015; Postles and Bartlett 2015). They also contribute to species inventories, but less is known about other types of biodiversity data they help generate. Bioblitzes may be beneficial for biodiversity monitoring by detecting rare and cryptic species, new invasive species to a region, range shifts in imperiled species, or help filling gaps in a

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species' geographic distribution. These metrics extend beyond what can be examined with species lists, including invasive species management targets or endangered species range expansions. In this way, bioblitzes may offer value to land managers, conservation agencies, and academic institutions. Therefore, it is critical to uncover the varied biodiversity utility that bioblitzes offer beyond simple species inventories for both land managers and conservation practitioners alike.

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Here, we evaluate the value of a bioblitz case study of a site (to generate biodiversity data, using the DeLuca Preserve, FL, USA), as a case study. First, we provide a replicable framework that can be expanded outside of our study site, with the aim of helping natural resource managers leverage bioblitzes to enhance land stewardship goals and empirical data analysis illustrating how this bioblitz has been useful in generating valuableadvance biodiversity data. We collection. We then use this framework to illustrate the scientific value of bioblitzesa bioblitz as a mechanism for generating biodiversity data, including baseline species inventories, novel species records, and indicators of regional biodiversity utility (Figure 1). In particular, we highlightOur work highlights how concentrated, short-term efforts can uncover locally novel records and contribute to site-level species inventories in ways that complement broader-scale biodiversity monitoring. We provide a replicable framework that can be expanded outside of our study site to either a broader scale or a different bioblitz region, with the aim of helping natural resource managers leverage bioblitzes to advance biodiversity generation.

Bioblitz biodiversity utility framework

We present a conceptual framework for evaluating the biodiversity value of bioblitzes by organizing metrics along two dimensions: their biodiversity utility for management and monitoring, and the analytical complexity required to calculate them (see Figure 1). The framework distinguishes site-level metrics that characterize biodiversity patterns within the site defined for the bioblitz (this includes, total observations, number of species, number of observers, and new species) from regional-level metrics that contextualize bioblitz observations relative to broader regional biodiversity data (this includes, relative frequency, relative rarity, and relative diversity of species, sampling intensity, and nearest external observation). Together, these metrics capture complementary dimensions of biodiversity value, ranging from baseline inventory completeness to the contribution of short-term, concentrated effort to regional biodiversity knowledge. Because we separate site-level and regional-level metrics, this framework provides a flexible tool that natural resource managers can adapt to evaluate the biodiversity outcomes of bioblitzes across different regions, taxa, and management objectives.

Methods

Study area

The DeLuca Preserve (27.~~67059~~267, -80.~~95230~~395) is situated in Osceola County in central Florida. The DeLuca Preserve is located near Yeehaw Junction, FL, a census-designated place in southern Osceola County about three hours south of the main University of Florida campus in Gainesville, FL, and about three hours north of the Miami metropolitan area in South Florida. The preserve was donated to the University of Florida by Elisabeth DeLuca in 2020 to place the land in permanent conservation (in collaboration with Ducks Unlimited) and to provide high-

quality habitat for student research. DeLuca Preserve is ~11,000 hectares and consists of pine flatwoods, Florida scrub, hardwood hammocks, forested and emergent wetlands, imperiled dry prairie, former citrus groves, and working cattle ranchland. The preserve hosts several state and federally listed species, such as the Florida grasshopper sparrow (*Ammodramus savannarum floridanus*), red-cockaded woodpecker (*Leuconotopicus borealis*), gopher tortoise (*Gopherus polyphemus*), and Florida panther (*Puma concolor couguar*). The area is now managed by the University of Florida Institute for Food and Agricultural Sciences (UF/IFAS) with varied management activities including active cattle ranching, hunting, prescribed burning in the pine flatwoods and pastures~~, and conserving waterfowl in the impounded wetlands on the property. The property has hunting leases~~.

The DeLuca Bioblitz

The DeLuca Bioblitz ~~began as an initiative from~~was initiated by the University of Florida Institute of Food and Agricultural Sciences (UF/IFAS) to promote public engagement with the newly established preserve while showcasing a conservation focal area of the Everglades Headwaters National Wildlife Refuge and Conservation Area. This annual event is designed as a free-ranging bioblitz, meaning participants are free to sample taxa and habitats of interest without any restrictions. However, we note that in the early years, there were some concerted efforts to obtain baseline biodiversity metrics with limited resources and people, which meant targeting specific portions or habitats of the property.

To date, the DeLuca Bioblitz has been running for four years and is expected to continue in future years. The dates of the previous bioblitzes were as follows: 26 February 2022, 24

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February 2023, 1 March 2024, and 7 March 2025. Over the course of these four years, participants have generated 13,161 total observations ~~from 1,252 species, with contributions from~~by 203 unique ~~individuals, hereafter referred to as~~ observers. With ~~its expanding datasets~~annual increases in total observations and ~~consistent public involvement~~unique observers, the DeLuca Bioblitz is positioned to play a long-term role in documenting local biodiversity trends and fostering awareness of the ecological significance of the DeLuca Preserve.

iNaturalist

iNaturalist is a citizen science platform where users can post and identify photographic or audio observations of any organism. It is accessible via the iNaturalist website or through their mobile application. As of October of 2025, iNaturalist has over 3 million observers and more than ~~2~~200 million observations. After an observation has been uploaded with a species identification on iNaturalist, other users can agree or disagree with the identification. Once an observation reaches a two-thirds community consensus, it is considered 'Research Grade'. ~~Research Grade~~All observations are made available for download on iNaturalist and Research Grade observations are uploaded to the Global Biodiversity Information Facility (GBIF), depending on license requirements, where they are made available to scientists around the world. The DeLuca Bioblitz used iNaturalist as the central tool to collect and engage with participants of the bioblitz, where only observations from iNaturalist were included in the bioblitz results. ~~We chose~~ iNaturalist was chosen as ~~our~~the bioblitz platform because it is free, has many users, is user-friendly, and the resulting data are increasingly used in ecological research (Mason et al., 2025b). At check-in,

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participants were asked if they were familiar with iNaturalist, and those who were unfamiliar with the platform were given a brief tutorial on how to download and engage with the app.

Analysis of records

We obtained all iNaturalist data for the DeLuca Bioblitz efforts by using the iNaturalist data export tool on July 1, 2025. ~~We aggregated the data~~Data is from the four different bioblitz efforts, described above, hereafter referred to as the DeLuca Bioblitz. For each species with a Research Grade observation(s), we also collected the total number of observations of that species in (a) Osceola County, (b) the state of Florida, and (c) the globe using the iNaturalist API (iNaturalist 2025). We report the relative frequency of these species found as part of the bioblitz efforts (site-level) relative to the total number of observations of each species in Osceola County, as well as the state of Florida~~.~~ (regional-level). Additionally, we used the iNaturalist API to obtain species origin information in relation to Florida (i.e., introduced, endemic) for all species recorded during DeLuca ~~Bioblitzes~~Bioblitz. To assess the extent to which the bioblitz filled in locations in species distributions, for every species with fewer than 100 observations from the DeLuca Bioblitz, we found the next closest Research Grade observation outside of DeLuca Preserve. To do this, we used all Research Grade iNaturalist data in Florida through June 27th, 2025, downloaded using the iNaturalist data export tool. While using the Global Biodiversity Information Facility (GBIF) is the recommended approach for obtaining Research Grade data, we noticed that the taxonomy did not fully align between iNaturalist-exported data and GBIF. Therefore, we chose to use iNaturalist data export, so our methods aligned with our methods for obtaining the DeLuca Preserve data.

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We also compared the cumulative spatial sampling effort from ~~all of~~ the DeLuca ~~Bioblitzes~~Bioblitz to other randomly distributed locations across Florida. To do this, we used the create_square function in the sf package (Pebesma and Bivand 2023; Pebesma 2018) and randomly generated 100 different polygons (i.e., square boundaries) in the state of Florida that matched the size of the DeLuca ~~preserve~~Preserve (Figure S1). In each of these randomly generated polygons, we gathered the number of observers, number of observations, and number of species documented at Research Grade level using the iNaturalist API. Comparisons between the DeLuca ~~Bioblitzes~~Bioblitz and the randomly generated polygons did not have the same temporal resolution. The DeLuca ~~Bioblitzes~~Bioblitz cumulatively represented data collected on ~~4~~four days from ~~4~~four separate years, whereas the randomly generated polygons included all-time records (i.e., there was no temporal filter applied to the polygon-generated data).

Data analysis and availability

All data analysis was conducted in R (R Core Team 2025) and relied heavily on the Tidyverse workflow (Wickham et al. 2019). All data and code are available in a Github repository (~~https://github.com/coreytcallaghan/DeLuca_bioblitz_analysis~~https://github.com/coreytcallaghan/DeLuca_bioblitz_analysis) and will be archived in Zenodo.

Results

~~*Overall contributions summary*~~

~~At~~ *Overview of biodiversity outcomes*

Across four annual bioblitzes, concentrated sampling at the DeLuca Preserve~~, we have hosted four bioblitzes, totaling 203 observers, 13,163 observations, and~~ generated a dense,

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taxonomically diverse dataset (N=892 ~~research grade~~Research Grade species~~. Throughout the four years, the number of unique observers has continued~~) that revealed numerous locally rare, novel, and under-documented species, while contributing disproportionately to ~~grow, with 59 observers in the first year, 29 new observers in the second year, 28 new observers in the third year, and 87 new observers in the last year. Only two observers were present at every single bioblitz thus far.~~regional biodiversity records. The ~~most commonly~~highest reported species were pink sundew (*Drosera capillaris;* 210 observations), fourpetal St. Johnswort (*Hypericum tetrapetalum;* 207 observations), shiny blueberry (*Vaccinium mrysinites;* 159 observations), saw palmetto (*Serenoa repens;* 155 observations), and orange milkwort (*Senega lutea*; 149 observations~~), with~~). A total 45% of Research Grade species ~~reported 1 or 2~~were documented two times or less, and 31% were only documented by a single observation. The spatial distribution of observations was heterogeneous across the preserve, with a pronounced concentration in areas of high accessibility (Figure 2). Species accumulation curves revealed a pronounced long-tailed distribution of species detection frequency and a non-asymptotic trajectory, indicating that species richness continues to increase with each sampling effort and that sampling completeness has not yet been achieved (Figure 3~~)~~), suggesting that more species will be found with ongoing efforts.

Using iNaturalist data to define species origin, we also examined the accumulation of Florida-endemic and introduced species across the four bioblitzes (Figure S2). Both groups exhibited upward trajectories consistent with the overall species accumulation curve, indicating that continued sampling will likely reveal additional taxa. To date, 69 introduced and 20 endemic species have been recorded; however, these counts are provisional and likely underestimate true

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totals due to incomplete sampling~~,~~ and ongoing data curation~~, and a lack of true coherence with iNaturalist information on species status~~.

~~*Gap filling*~~

Comparison with random landscape benchmarks

When comparing with randomly-sampled polygons, we found that the efforts associated with the DeLuca Bioblitz led to markedly higher observations per km² (118.0 vs. 20.1 ± 44.7; $t_{99} = -21.9$, $p < 0.001$) and species per km² (7.98 vs. 4.28 ± 3.77; $t_{99} = -9.80$, $p < 0.001$) than the statewide average, while observers per km² did not differ significantly (1.82 vs. 2.40 ± 3.79; $t_{99} = 1.52$, $p = 0.13$) (Figure 4). Shannon's diversity index further confirmed high diversity of observations at the preserve (H' = 5.81), approaching statewide diversity (H' = 6.82), indicating that the DeLuca Bioblitz captured both common and rare taxa in a relatively short time frame compared to the entire time frame for the 100 random polygons.

For species with less than 100 observations recorded during the DeLuca ~~Bioblitzes~~Bioblitz, we found that the average median linear nearest distance of the species recorded outside of the DeLuca ~~preserve~~Preserve was 7.043 km and ranged from 0.0035 to 7,595 km. The nearest observations represent common species such as white beggarticks (*Bidens alba*), ballmoss (*Tillandsia recurvata*), and Pickerelweed (*Pontederia cordata*). The species that was furthest away was false smut of palm (*Graphiola phoenicis*), a fungal leaf disease that affects palm species, where the nearest observation was in Lisbon, Portugal. Though there are no other Research Grade iNaturalist observations in Florida, this species has been known to occur in this region (Elliott 2005). Upon speaking with an expert in this field, they agree with the genus-level

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identification of this species, *Graphiola*, but did not agree that species-level identification is possible based on photographic evidence, given that there are multiple species of *Graphiola* worldwide (Piepenbring et al. 2012). Nevertheless, this is still the first observation of *Graphiola* on iNaturalist in the United States.

An analysis of relative rarity

Comparison with county- and state-level biodiversity baselines revealed that the DeLuca ~~Bioblitzes~~Bioblitz contributed disproportionately to records of regionally rare and infrequently documented species. A subset of species recorded at DeLuca (2 species: *Condylostylus flavipes* and *Aethes triassumenta*) were observed only within the preserve and had not been documented elsewhere in Florida (Figure 5). Bioblitz-derived observations accounted for a disproportionately high share of county-level occurrence records, particularly within Osceola County, where many species were represented primarily or exclusively by DeLuca ~~observations (Figure S3). Shannon's diversity index further confirmed high diversity of observations at the preserve (H' = 5.81), approaching statewide diversity (H' = 6.82), indicating that the bioblitzes captured both common and rare taxa in a relatively short time frame compared to the entire time frame for the 100 random polygons.~~Bioblitz observations (Figure S3).

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Discussion

Bioblitzes are increasingly popular events (Figure S4), concomitant with a general increase in participatory citizen science in general (Feldman et al. 2021). While there are many benefits of such events for both engagement and natural resource management (Mason et al. 2025b), we

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provide a framework and an assortment of biodiversity metrics that can be used to evaluate the utility of bioblitzes for collecting relevant biodiversity data. Using a four-year case study, we show that a concentrated bioblitz effort can (1) generate dense, taxonomically diverse datasets over short time frames, (2) detect rare, endemic, and introduced species of management relevance, and (3) contribute disproportionately to regional biodiversity records. We evaluate potentially valuable data both for local, place-based adaptive management of natural resources, where, in this instance, observers found at least 33 endemic species and 71 introduced species. We also illustrate how data from a bioblitz can contribute to broad, ~~landscape~~regional-scale knowledge about biodiversity, ~~where we found that bioblitzes contribute relatively~~with biodiversity metrics from the ~~same~~DeLuca Bioblitz matching or ~~greater knowledge in a 96-hour time frame as random~~exceeding those from randomly selected polygons ~~throughout~~across Florida ~~during their entire iNaturalist history.~~despite the much shorter sampling duration. Overall, ~~for the first time, we show how a~~ our results demonstrate that concentrated ~~effort organized through a~~ bioblitz efforts can ~~contribute~~generate essential ~~data~~biodiversity information for natural resource management.

The value of a given biodiversity observation is context-dependent. Biodiversity observations may be locally common species, new county occurrences, threatened species, or the first detection of a potential invader. However, we note that there are often context-dependent qualitative findings, where a single record of a specific species can be very valuable, such as a finding of a non-native species (Yeshwanth et al. 2025) or of a species of conservation concern (Wilson et al. 2020; Mesaglio et al. 2021). The DeLuca Bioblitz uncovered several notable findings, including a critically imperiled species, the Florida black dark bee (*Stelis ater*), along

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with multiple threatened (e.g., *Hemiargus ceraunus*) and endemic species (e.g., *Lupinus floridanus*), underscoring the conservation value of this property and the importance of continued biodiversity monitoring through the bioblitz. Since bioblitzes concentrate effort in space and time, they are especially powerful in this regard, acting as early-detection systems that complement ongoing monitoring programs. Further, when enough observations accumulate at a local site, they provide the foundation for mapping biodiversity patterns with increasing resolution (e.g., Figure 1). As analytical approaches for citizen science data continue to advance (Beninde et al. 2023; Stewart et al. 2023), such datasets could allow researchers not only to document presence but also to estimate relative abundance and detect temporal trends. In this way, repeated and well-organized bioblitzes could form the basis of long-term biodiversity monitoring at the site scale.

At a broader scale, we situated the DeLuca Bioblitz efforts in a statewide context by leveraging a random polygon approach to ~~compare the biodiversity data generated from a short-temporal scale bioblitz to all collected temporal data documented on iNaturalist around the state of Florida. This comparison highlights the distinctive value of the concentrated effort of a bioblitz. Our random polygon analysis (Figure 4; Figure S1) provides a novel way to benchmark bioblitz contributions against broader iNaturalist activity. These comparisons illustrate the intensity of participation these events can generate.~~benchmark the biodiversity data generated from a short-temporal scale bioblitz. In terms of observations and species per unit area, the DeLuca Bioblitz efforts contributed 487% greater observations per km² and 86.5% greater species per km² than the mean of random polygons, despite the dramatically shorter time frame. ~~This illustrates that bioblitzes can produce unusually dense datasets, creating snapshots of biodiversity that provide a~~

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~~complementary and powerful perspective.~~Our random polygon analysis (Figure 4; Figure S1) provides a way to benchmark bioblitz contributions against broader iNaturalist activity. However, we acknowledge that underlying biodiversity, habitat composition, and accessibility vary across polygons, and observer behavior on iNaturalist is not spatially random, all of which can contribute to variation among polygons. As such, these comparisons are best interpreted as coarse contextual benchmarks rather than direct ecological contrasts. Nevertheless, this analysis illustrates that bioblitzes can produce unusually dense datasets, creating snapshots of biodiversity that provide a complementary and powerful perspective. We further conducted an exploratory analysis, comparing DeLuca Bioblitz data to iNaturalist data from 39 Florida state parks (Lowe et al., 2026), to evaluate active versus passive sampling (Figure S5). Despite the DeLuca Bioblitz data being collected over just four days, compared to the full temporal extent of the state park data, the bioblitz yielded more observations and species than expected based on state park trends. This illustrates the value of bioblitzes in generating more observations and species than passive sampling. Given that bioblitzes can generate robust species inventories in 24 hours that rival entire temporal scales in our random polygons, land managers should consider the benefits of these non-traditional sampling events to create baseline species inventories. Lastly, conservation practitioners interested in both rare and potentially invasive species can leverage bioblitz events to document large amounts of data rapidly, compared to more traditional monitoring methods that can be more costly and time-consuming.

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Our evaluation focused on quantitative metrics that are broadly applicable across bioblitz events, but this represents only one dimension of their value. Future work could strengthen these insights by directly comparing bioblitz outcomes with professionally collected survey data, assessing

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how well volunteer-collected records align with standardized monitoring (Katzer et al. 2025). Additional approaches might include examining taxonomic biases (e.g., toward charismatic or easily photographed taxa or taxonomic uncertainty), quantifying participant learning or engagement outcomes (Echeverria et al. 2021; Niemiller et al. 2021), or integrating bioblitz data with other monitoring tools such as eDNA or remote sensing (Granqvist et al. 2025; Cavender-Bares et al. 2022). Importantly, the results we present here emphasize short-term outcomes—the concentrated biodiversity snapshots generated over the course of a single event. The longer-term value, particularly the potential for repeated bioblitzes to build cumulative datasets that enable site-level monitoring and adaptive management, remains to be fully realized. As such, bioblitzes represent not only a means to engage the public but also an underutilized mechanism for generating robust, repeatable biodiversity data with clear relevance for conservation and management.

Conclusions

We provide an overview of how a bioblitz can provide valuable biodiversity data. While we introduce quantitative metrics and provide a conceptual framework (Figure 1) for how data from a bioblitz can contribute to both local place-based natural resource management as well as ~~landscape~~regional-scale biodiversity research. However, we stress that hosting a bioblitz is time-consuming and requires logistical planning (Tiago et al. 2024). Nevertheless, based on our results and collective experience, we suggest that natural resource managers should consider the use of bioblitzes and how they can contribute to adaptive management goals.

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We thank the participants of the DeLuca Bioblitz and all iNaturalist contributors, observers and identifiers, for their valuable observations and species identifications, which made this research possible. We also thank those who assisted with event coordination and execution. We thank Braham Dhillon for identification help. CTC acknowledges that this research was supported in part by the intramural research program of the U.S. Department of Agriculture, Hatch, FLA-FTL-006297. This work is supported by the McIntire-Stennis program, project award no.FLA-FTL-006659, from the U.S. Department of Agriculture's National Institute of Food and Agriculture.

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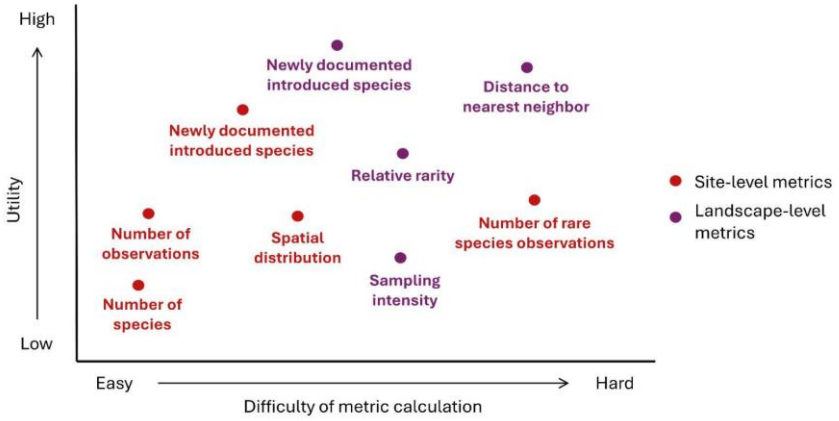
Figures and Tables

(A) Conceptual diagram



(B) Site-level metric definitions

Metric	Definition
Number of observations	Count of Research Grade observations in the site.
Number of species	Count of uniquely documented species in the site.
Spatial distribution	Density heat map of observations in the site.
Newly documented introduced species	List of non-native species that are documented at the site for the first time.
Number of rare species observations	List of species that are rarely documented in the site.

(C) Landscape-level metric definitions

Metric	Definition
Sampling intensity	Comparison of site-level observation and species counts to 100 random polygons in the region.
Distance to nearest neighbor	Distance of each site-level observation with less than 100 observation to nearest observation outside of site.
Newly documented introduced species	List of non-native species newly documented in the region.
Relative rarity	Comparison of site-level observation count to the observation count in the region.

Table 1. Metrics used to assess bioblitz biodiversity utility. Metric scale is defined as site-level (SL) which characterizes biodiversity patterns within the site defined for the bioblitz and regional-level (RL) which contextualizes bioblitz observations relative to broader regional biodiversity data. We additionally provide a brief definition of each metric and the methodology used to calculate them.

Metric	Scale	Definition	Calculation
Total observations	SL	Total observations taken at the site.	Count of Research Grade observations.

	RL	The relative number of observations at the site compared to the surrounding region.	One hundred random polygons that were the size of the site were generated within the region. For each polygon, the number of observations was calculated and compared to the corresponding values from the site.
Number of observers	SL	Unique observers who submitted observations at the site.	Count of unique observers from Research Grade observations.
	RL	The relative number of observers at the site compared to the surrounding region.	Follows same methodology as total observations (RL) but compares observers.
Number of species	SL	Number of unique species observed at the site.	Count of unique species from Research Grade observations.
	RL	The relative number of unique species at the site compared to the surrounding region.	Follows same methodology as total observations (RL) but compares number of unique species.
New species	SL	Measure of new species at the site based on sampling effort.	Species accumulation curve.
Relative frequency	RL	Frequency of species at the site relative to the region.	For each species, proportion of observations at the site is divided by its proportion of observations in the region.
Relative rarity	RL	Measure of species rarity based on number of observations at the site and region.	Relative frequency is used to define categories for rarity, including site-only, locally rare, common everywhere, rare everywhere, and underreported everywhere.
Relative diversity of species	RL	Diversity of species at the site compared to the region.	Shannon's diversity index.
Nearest external observation	RL	Distance from the site to the nearest external observation for species with fewer than 100 site-level observations.	For each qualifying species, the minimum distance to any observation outside the site boundary was computed.

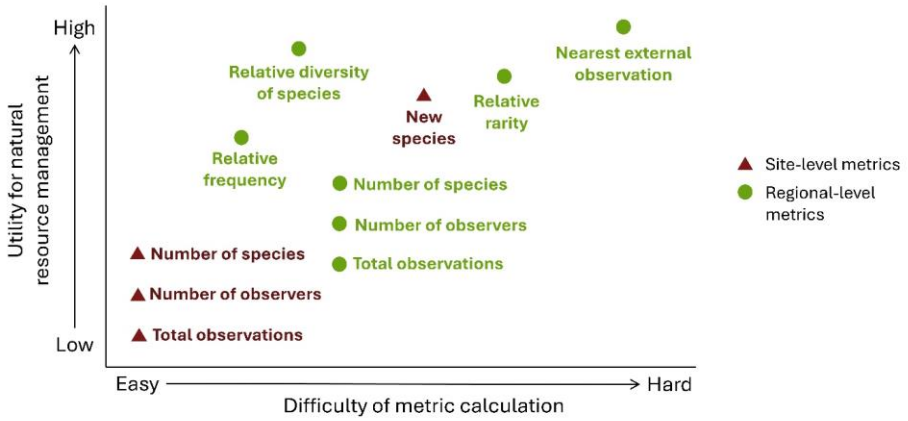


Figure 1. ~~(A)~~ Conceptual diagram showing the utility and difficulty of calculating the ~~nine~~bioblitz biodiversity utility metrics explored in this paper. Metrics are grouped as either site-level (defined relative to the bioblitz site) or ~~landscape~~regional-level (defined as the contribution of the bioblitz site to the broader region). ~~(B) Definitions~~See Table 1 for definition of ~~site-level metrics based on Research Grade iNaturalist observations within the bioblitz site. (C) Definitions of landscape-level metrics, which incorporate Research Grade iNaturalist observations from outside the bioblitz site to represent the broader region~~each metric.

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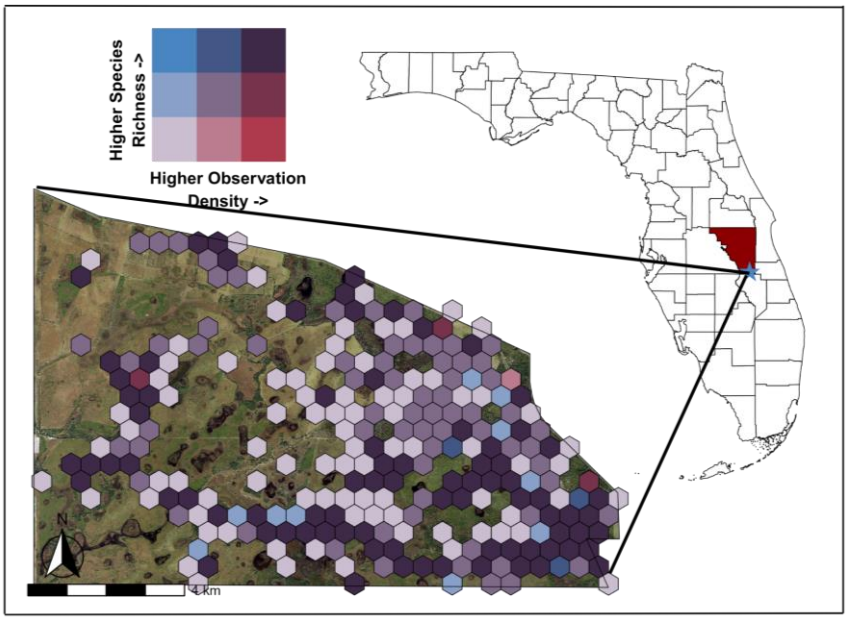

Figure 2. A map of DeLuca Preserve, showing its location in Florida and its relative remoteness. Osceola County, where the preserve is located, is highlighted in red. Additionally displayed is a density heat map of where the observations within the preserve have been derived from the DeLuca Bioblitzes, and the concentration of observations and species richness. Bivariate colors reflect quantile-based classification (dim = 3) of observation density and species richness.

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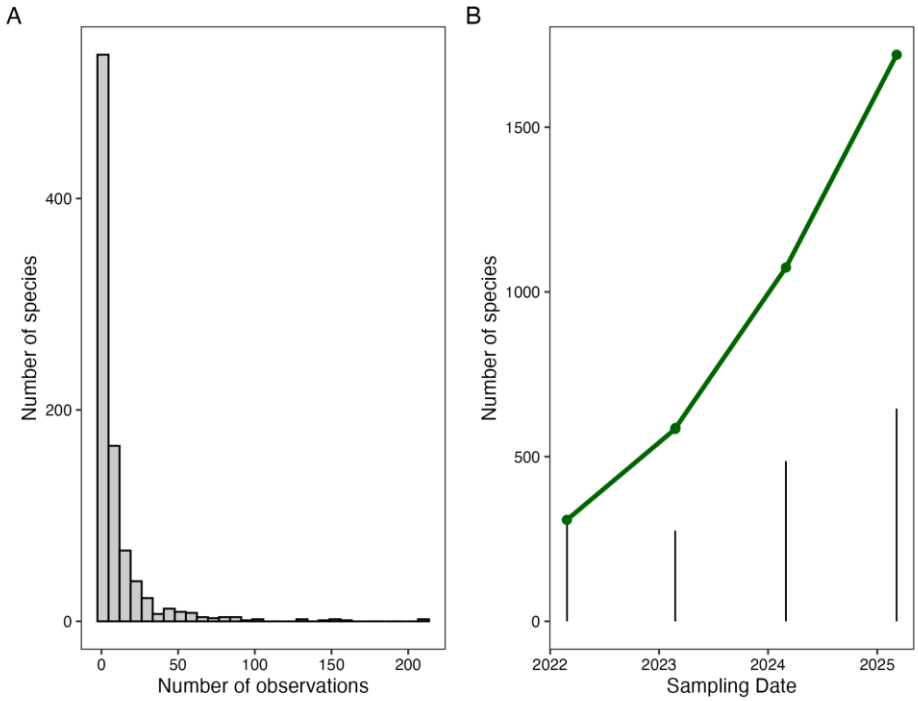
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Number of species
400
200
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B
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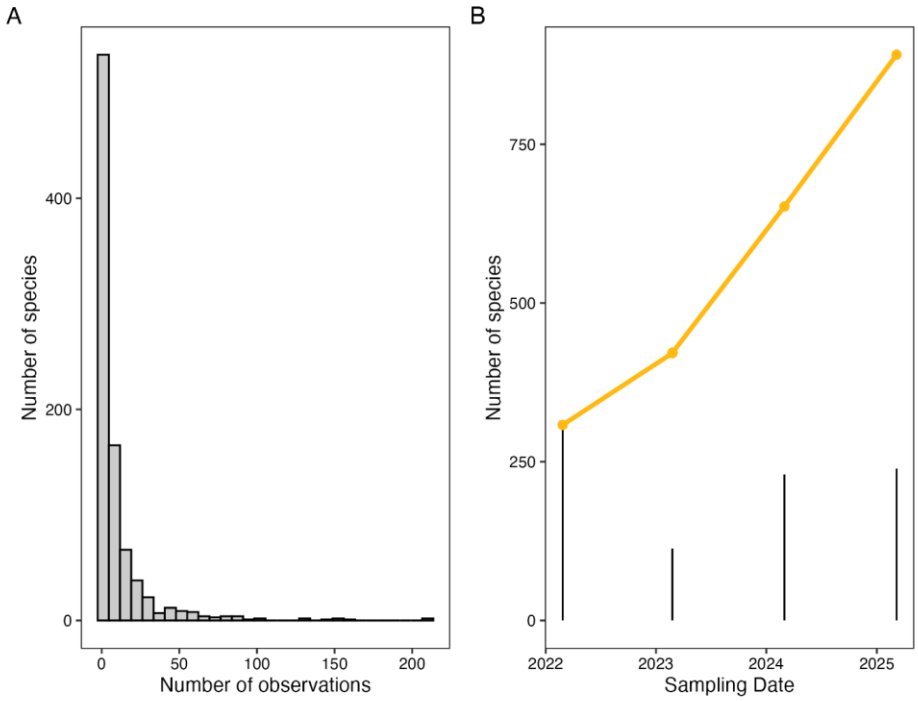

Figure 3. (A) Frequency distribution of species detection rates across the DeLuca Bioblitz dataset, showing the number of species in each observation group. (B) Species accumulation across sequential bioblitz events (2022–2025), indicating continued discovery of novel taxa and incomplete sampling saturation. The vertical lines represent the number of new Research Grade species found that year.

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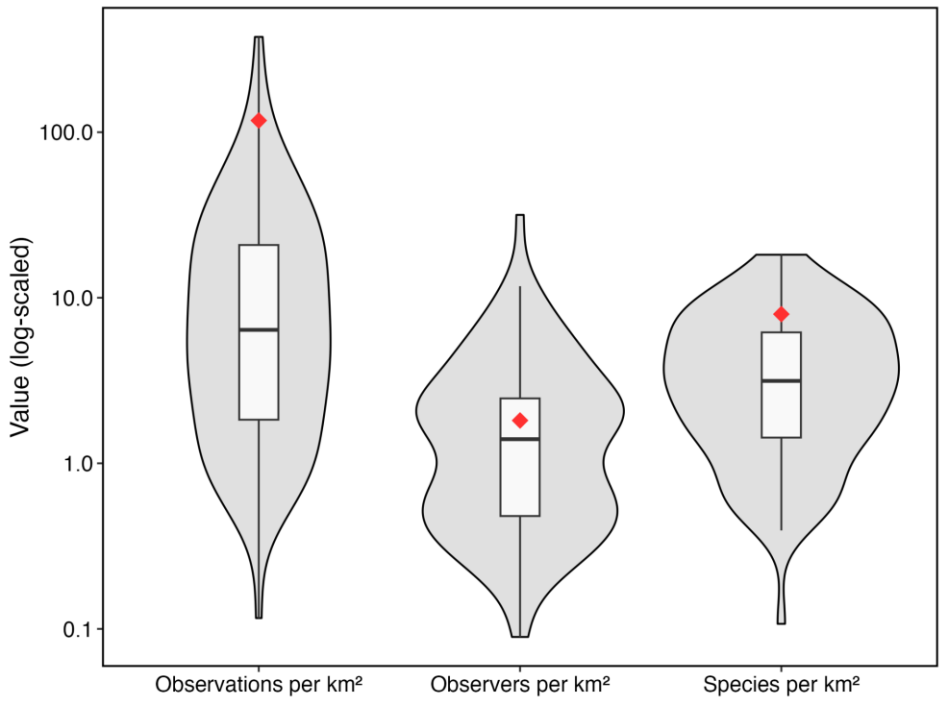
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Observers per km²
Species per km²

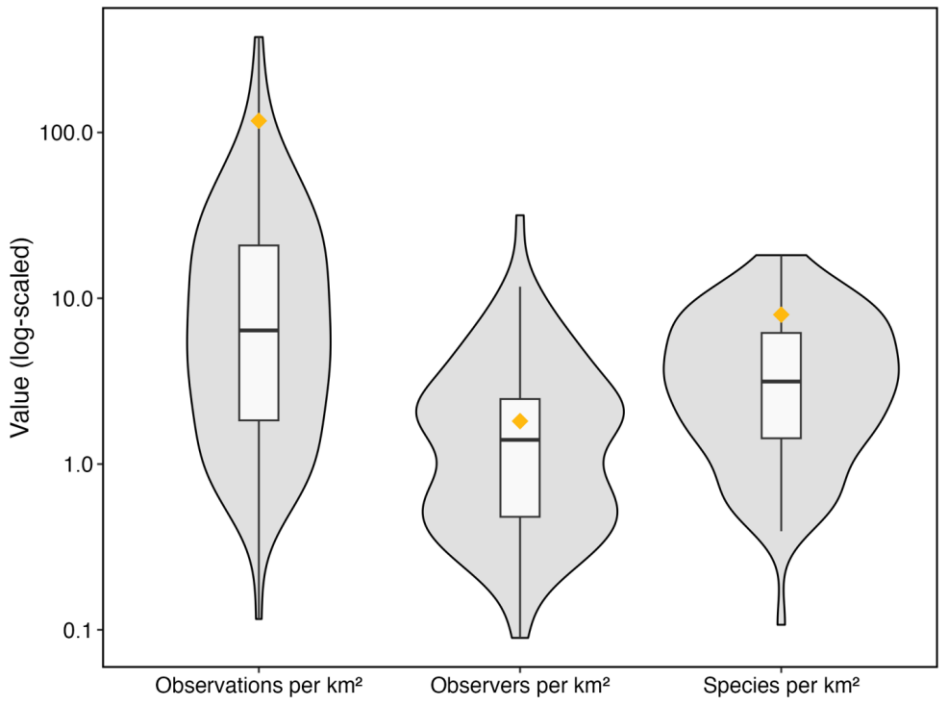

Figure 4. Comparison of biodiversity metrics at the DeLuca Preserve relative to 100 randomly selected polygons of equivalent area across Florida. Violin and boxplots represent the statewide distribution of iNaturalist data in random polygons for three metrics normalized by area (observers, observations, and species per km²), while the diamond markers indicate values from the DeLuca Bioblitz dataset. DeLuca exhibited markedly higher observation density (118.0 vs. 20.1 ± 44.7; $t_{99} = -21.9$, $p < 0.001$) and species density (7.98 vs. 4.28 ± 3.77; $t_{99} = -9.80$, $p < 0.001$) than the statewide average, whereas observer density did not differ significantly (1.82 vs. 2.40 ± 3.79; $t_{99} = 1.52$, $p = 0.13$).

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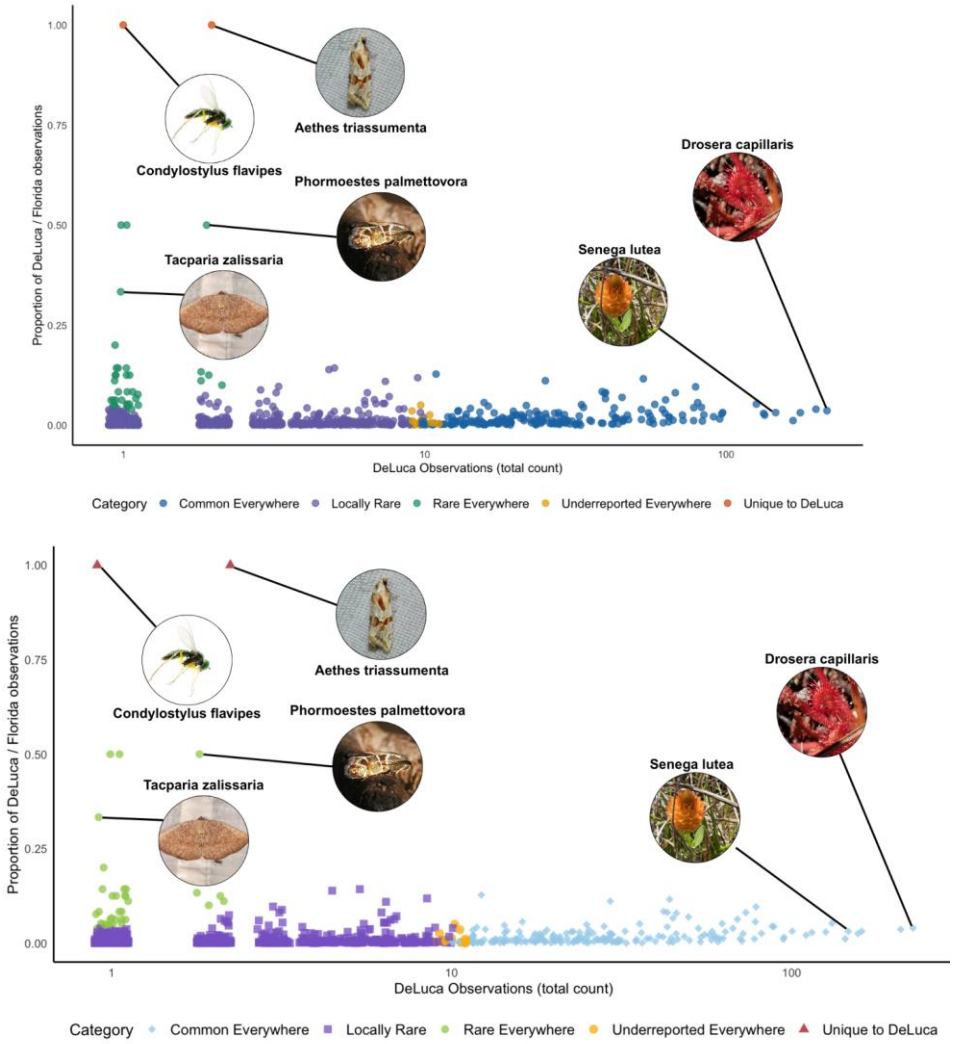

Figure 5. Each point represents a species observed at DeLuca, plotted by its total observation count at DeLuca (log-scale) versus its proportional representation relative to Florida-wide observations. Colors indicate rarity categories: blue represents common everywhere (> 10 observations in DeLuca; > 25 observations in Florida), yellow represents underreported everywhere (= 10 observations in DeLuca; <= 25 observations in Florida), purple represents locally rare (< 10 observations in DeLuca; > 25 observations in Florida), green represents rare everywhere (< 10 observations in DeLuca; < 25 observations in Florida); and red represents

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unique to DeLuca. Image attributions: Odin Toness (*Condylostylus flavipes*), Laura Gaudette (*Tacparia zalissaria*), Jeff Niznik (*Aethes triassumenta*), Liz Childress (*Phormoestes palmettovora),* Aurora Alcaide (*Senega lutea*), and Donald Fraser (*Drosera capillaris*).